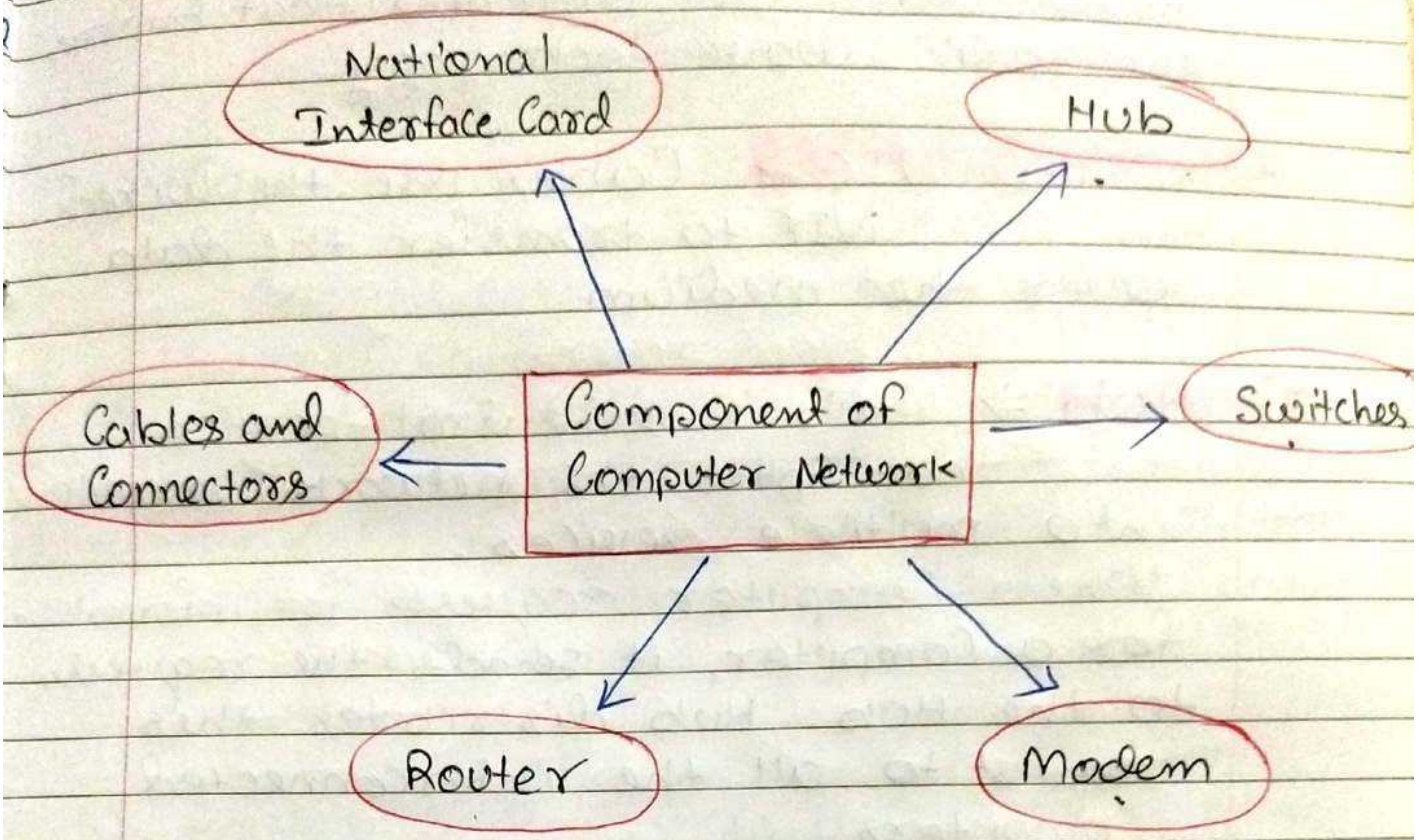
Computer Network

Computer Network ⇒ • Computer Network refers to interconnected Computing devices that can exchange data and share resources with each other.

- Computer Network is a group of Computers Connected with each other through wires, optical fibres or optical links so that various devices can interact with each other through a network.
- A Computer Network is a System in which multiple Computers are connected to each other to share information and resources.
- A Computer network is a set of Computers sharing resources located on or provided by Network nodes. The Computers use common Communication protocols over digital interconnections to Communicate with each other.
- A Computer Network is defined as a system that connets two or more Computing devices for transmitting and sharing information.

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Components of Computer Network:



1) NIC (National Interface Card) ⇒

NIC is a device that helps the computer to communicate with another device.

The network interface card contains the hardware addresses, the data-Link Layer protocol use this address to identify the System on the network so that it transfers the data to the correct destination.

There are two types of NIC.

1) **Wireless NIC** ⇒ All the modern Laptops use the wireless NIC.

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in wireless NIC, a connection is made using the antenna that employs the radio wave technology.

2) Wired NIC ⇒ Cables use the wired NIC to transfer the data over the medium.

2) Hub ⇒ Hub is a central device that splits the network connection into multiple devices.
When computer requests for information for a computer, it sends the requests to the Hub. Hub distributes this request to all the interconnected computers.

3) Switches ⇒ Switch is a networking device that groups all the devices over the network to transfer the data to another device.
A switch is better than Hub as it does not broadcast the message over the network. Such as it sends the message to the device for which is belongs to.

4) Cables and Connectors ⇒ Cables is a transmission media that transmits the communication signals.

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There are three types of cables:

1) Twisted pair cable ⇒ It is a high-speed cable that transmits the data over 1GbPS or more.

2) Coaxial cable ⇒ Coaxial cable resembles like a TV installation cable.
Coaxial cable is more expensive than Twisted pair cable.

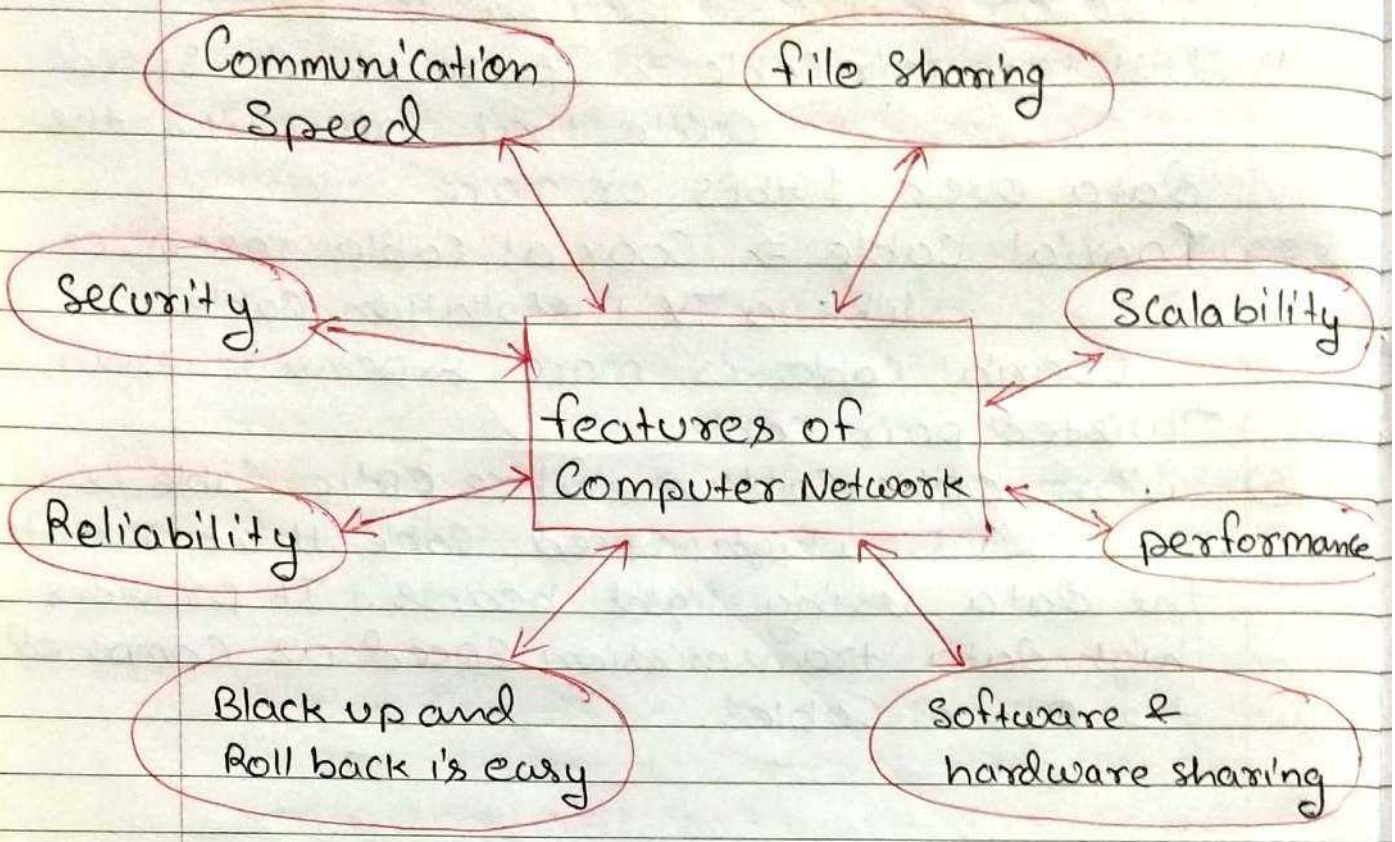
3) fibre optic cable ⇒ fibre optic cable is a high-speed cable that transmits the data using light beams. It provides high data transmission speed as compared to other cables.

5) Router ⇒ Router is a device that connects the LAN to the internet. the router is mainly used to connect the distinct networks or connect the internet to multiple Computers.

6) Modem ⇒ Modem Connects the Computer to the internet over the Existing telephone line.
A modem is not integrated with the Computer motherboard.
A modem is a separate part on the PC Slot found on the motherboard.

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features of Computer network



1) **Communication Speed** ⇒ Network provides us to Communicate over the network in a fast and efficient manner.
for Ex ⇒ we can do video Conferencing, email messaging etc. over the internet. The Computer network is a great way to share our knowledges and ideas.

2) **file sharing** ⇒ One of the most important features of a Computer Network is that with their help you can Easily share the files Between Different Systems that are Connected with Each other

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through transmission media.

3) **Scalability** ⇒ The Scalability of a Computer network simply means that we can add new nodes or components of the network easily. Any computer network must be scalable so we can extend it easily by adding new nodes. After adding new nodes to the network the speed of connection decreases which leads to a decrease in the speed of transmission of data.

4) **Performance** ⇒ The performance of a Computer network is measured using response time and with the help of the speed of data transmission. For better performance the response time of sending and receiving data from one node to another should be minimum.

5) **Software and hardware sharing** ⇒ We can install the application on the main server, therefore the user can access the applications centrally. So, we do not need to install the software on every machine, similarly, hardware can also be shared.

6) **Back up and Roll back is easy** ⇒ Since the files

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are stored in the main server which is centrally located. Therefore, it is easy to take the back up from the main server.

7) **Reliability** ⇒ With the help of Computer Networks there are fewer chances for the occurrence of failure and in case if there is any failure then recovery is fast.

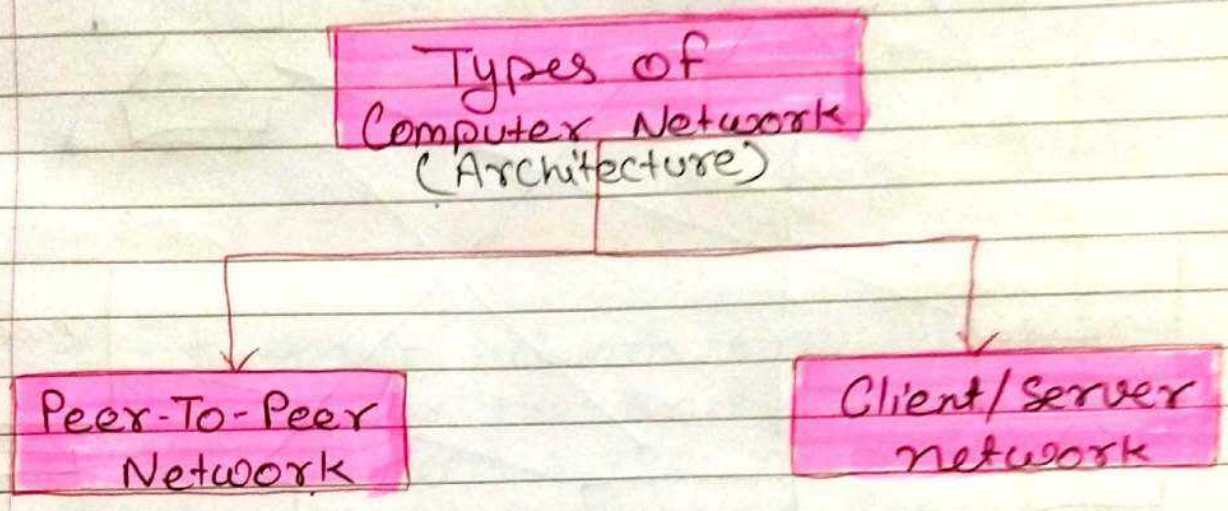
8) **Security** ⇒ Security is one of the main characteristics of Computer Networks, Thus a computer network should be secure so that the data transmitting over the network should be safe from any unauthorized access, the data send by the sending node should be received as it is at the receiving node, which simply means there should be no loss of data during the transmission of the data.

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Computer Network Architecture

Computer Network Architecture is defined as the physical and logical design of the Software, hardware, protocols and media of transmission of data.

There are two types of network Architectures.

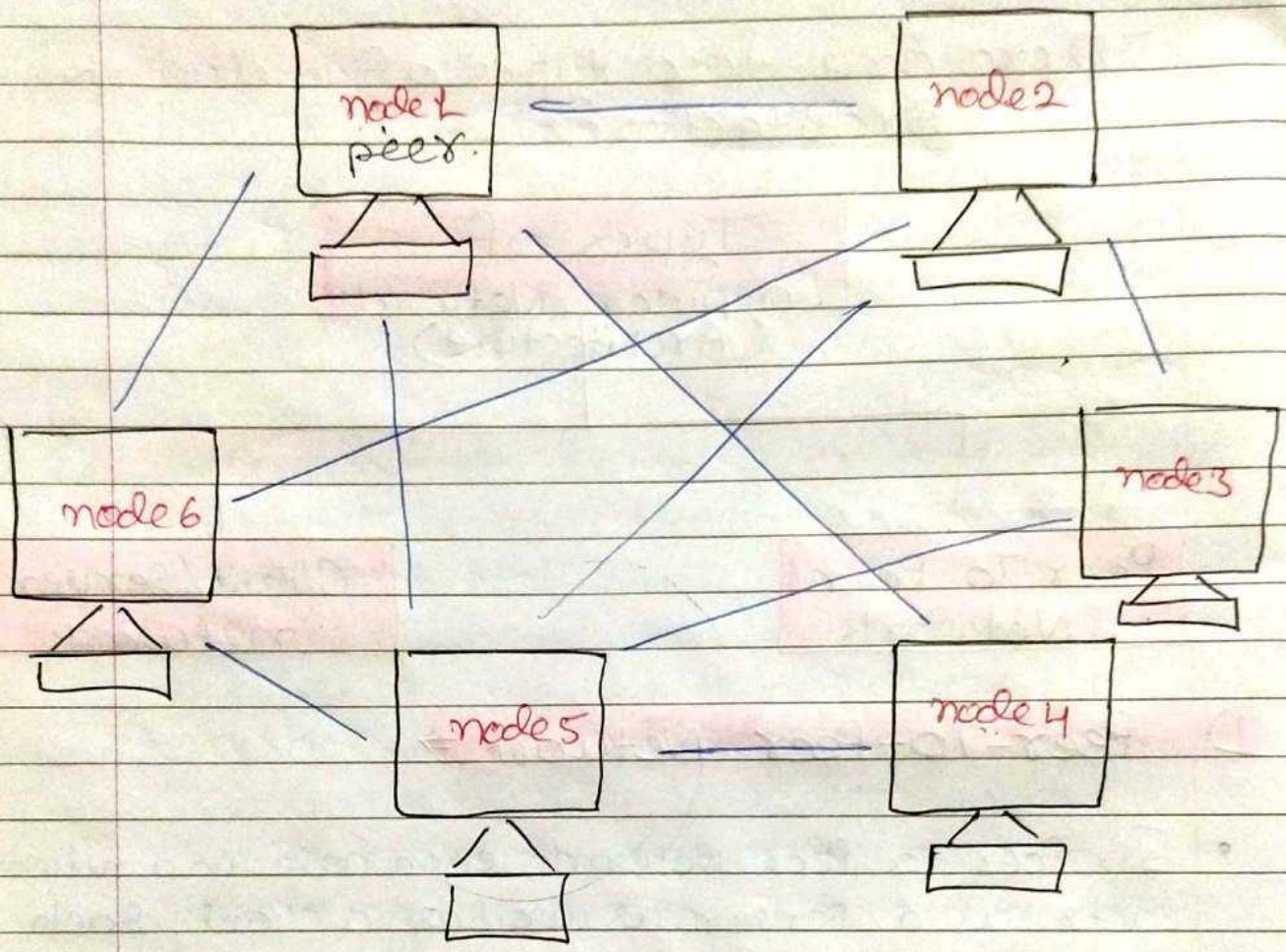


1) Peer-To-Peer Network ⇒

- In Peer to Peer Network, each node in a network acts as a server as well as a client. Each node on network is known as a peer.
- There are no dedicated servers required to provide services to the clients.
- Peer-to-Peer network is useful for small environments, usually upto 10 Computers.
- Special permissions are assigned to each Computer for sharing the resources, but this can lead to a problem if the Computer with the resource is down.

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- Each node is capable of requesting services and can also provide the server to other nodes.



Advantages of Peer-To-Peer Network:

- It is less costly as it does not contain any dedicated server.
- if one computer stops working but, other computers will not stop working.
- It is easy to set up and maintain as each computer manages itself.

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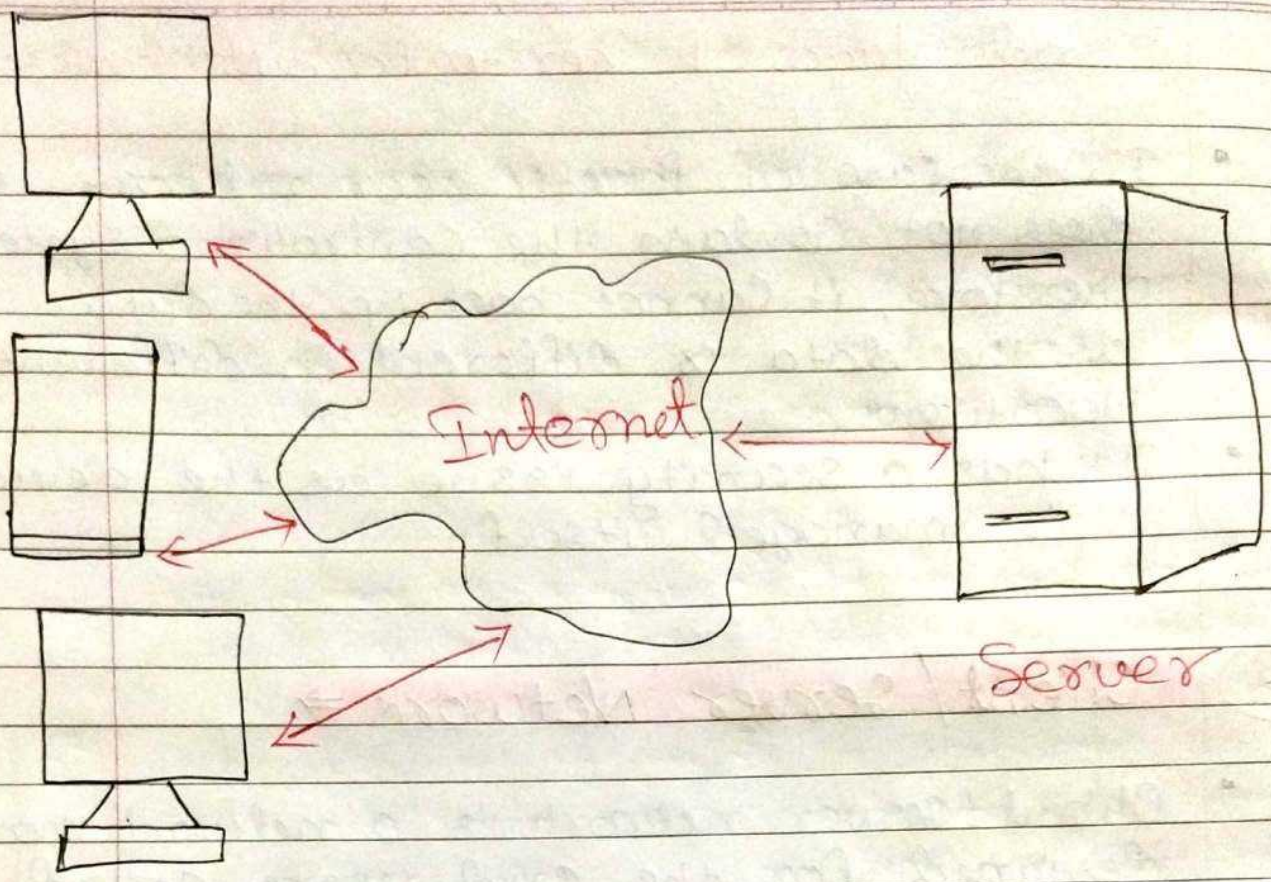
Disadvantages of Peer-to-Peer Network ⇒

- In the case of Peer-to-Peer network, it does not contain the centralized system. Therefore, it cannot back up the data as the data is different in different locations.
- It has a security issue as the device is managed itself.

2) Client/Server Network ⇒

- Client/server network is a network model designed for the end users called Clients, to access the resources such as songs, video etc. from a central computer known as server.
- The central controller is known as a server while all other computers in the network are called clients.
- A server performs all the major operations such as security and network management.
- All the clients communicate with each other through a server.
- Suitable for a larger network.

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Advantages of Client/server network:

- A Client/server network Contains the Centralized System. Therefore we can backup the data easily.
- A Client/server network has a dedicated server that improves the overall performance of the whole system.
- Security is better in client/server network as a single server administers the shared resources.
- It also increases the speed of the sharing resources.

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Disadvantages of Client/server Network:

- Client/server network is expensive as it requires the server with large memory.
- A server has a Network Operating System (NOS) to provide the resources to the clients, but the cost of NOS is very high.
- It requires a dedicated network administrator to manage all the resources

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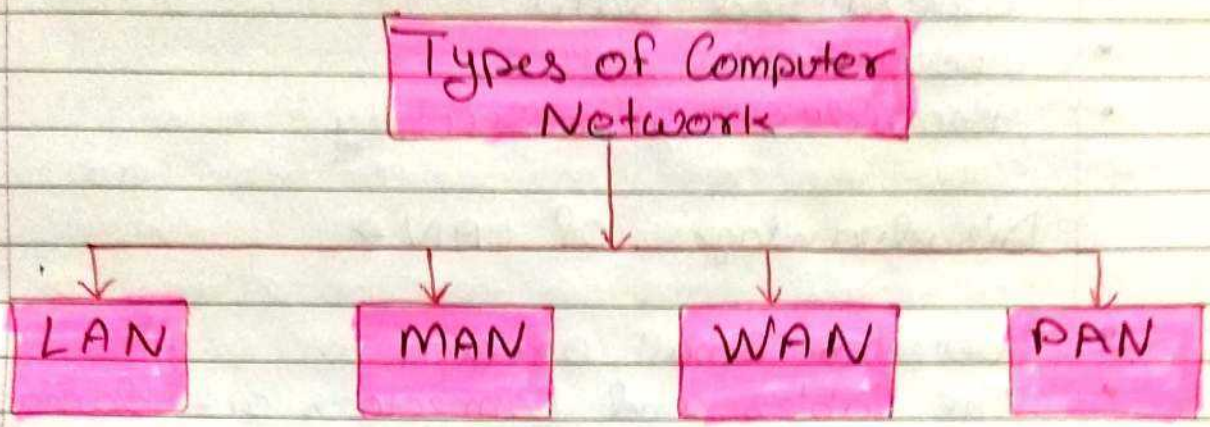
Difference between Peer-to-Peer and Client server network.

Peer-to-Peer Network	Client-server Network
1) The peer-to-peer network model is distributed and decentralized nature.	1) The Client-server network model is also distributed in nature but it is Centralized.
2) Peer-to-Peer Network focuses on Connectivity.	2) It focuses on information sharing.
3) In peer-to-peer Network, Clients and Server are not differentiated.	3) In Client-server Network, Clients and server are differentiated, Specific server and Clients are present.
4) Peer-to-Peer Network are less Stable if number of peer is increase.	4) Client-Server Network are more Stable than Peer-to-Peer Network.
5) Peer-to-Peer Network Each peer has its own data	5) Client-server Network, Centralized server is used to store the data.
6) Peer-to-Peer is less Expensive to implement	6) Client-Server network is more Expensive to implement.

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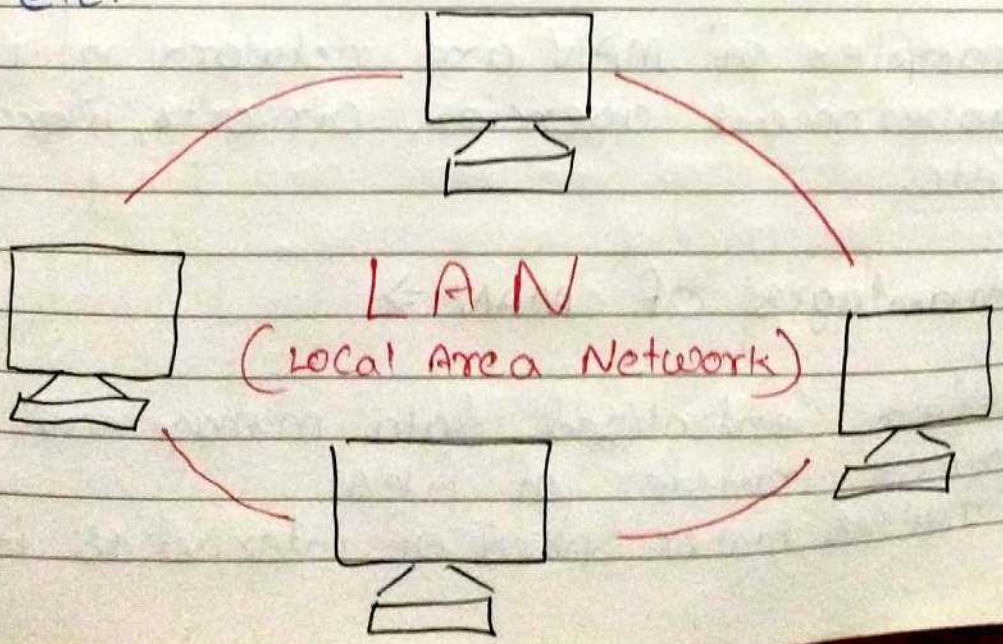
Types of Computer Network

A Computer network can be categorized by their size. A Computer Network is mainly of four types:



1) **LAN** (Local Area Network) ⇒

- LAN is a group of computers connected to each other in a small area such as building, office, schools etc.
- LAN is used to connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.



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Advantages of LAN ⇒

- High data transfer rate
- Ease of setup
- Centralized data
- Low cost
- Provides higher security

Disadvantages of LAN ⇒

- Covers small area
- The cables and connectors get damaged easily.
- Requires administrative time.

2) MAN (Metropolitan Area Network) ⇒

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a large Network.

- Examples of MAN are networking in government agencies, airports, libraries etc.

Advantages of MAN ⇒

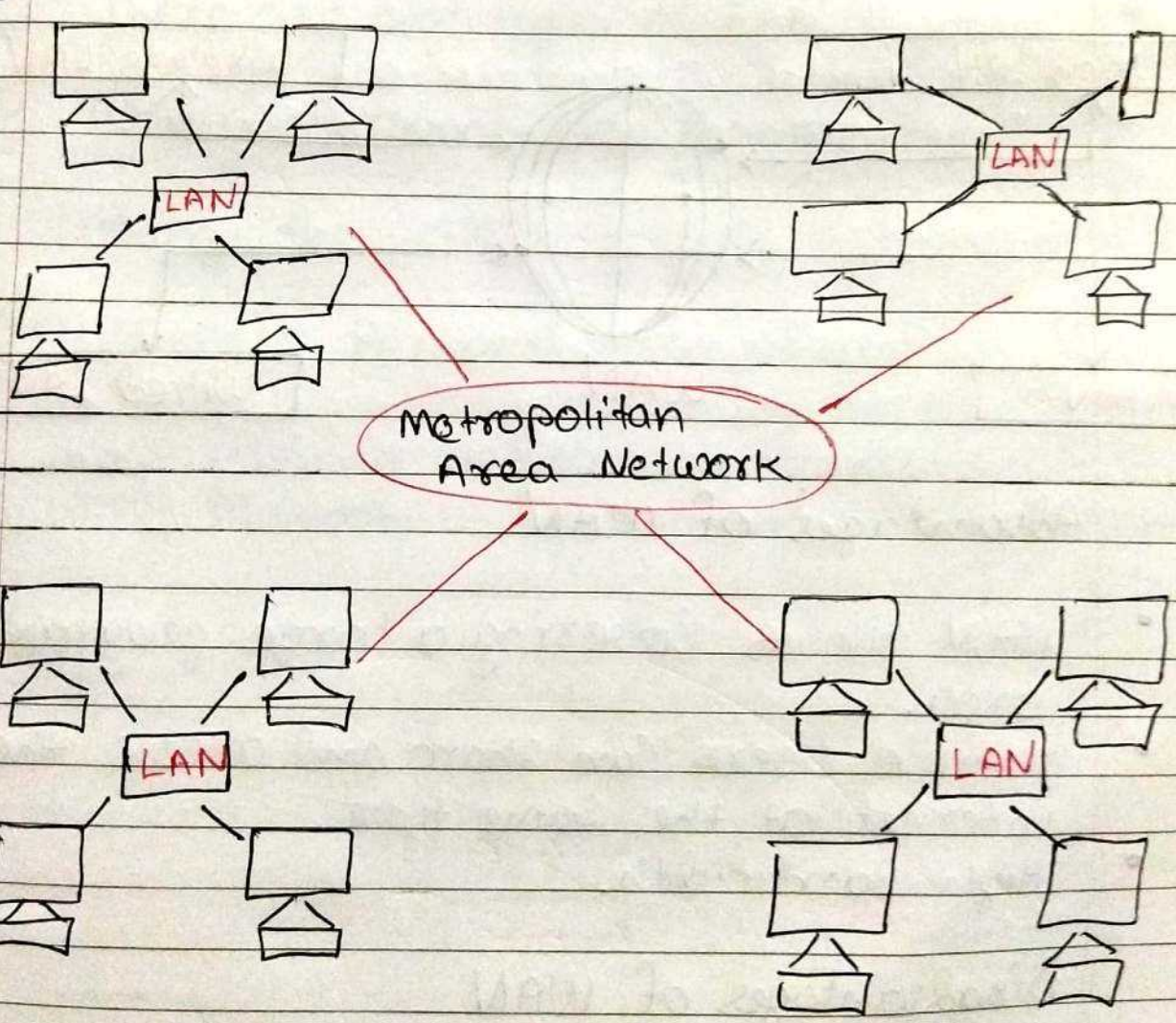
- offers Centralized data management.
- Quick transfer of files
- Provides higher speed of internet as it

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Uses fiber optics.

Disadvantages of MAN ⇒

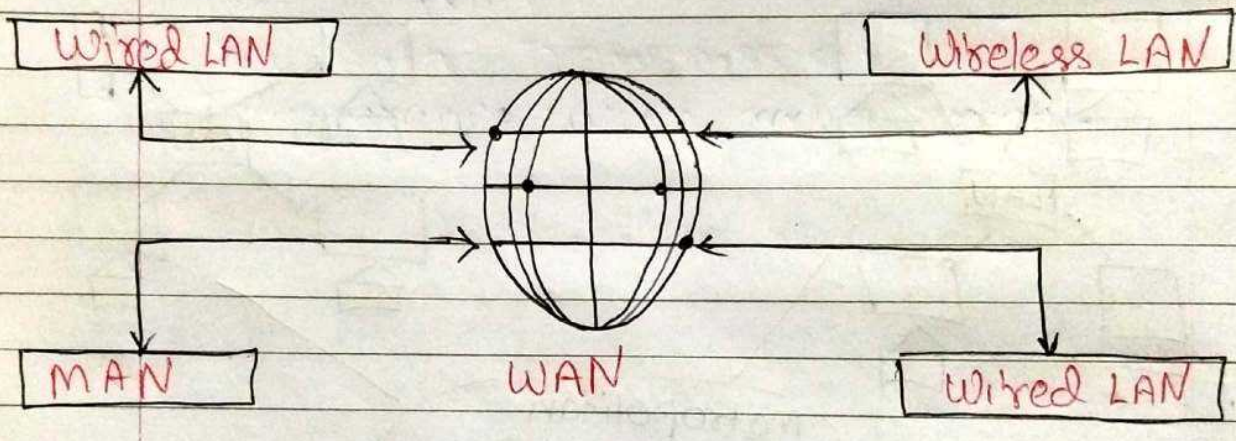
- Difficult to handle due to large network size.
- Risk of hacking
- High installation cost as it requires fiber optics.



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3) WAN (Wide Area Network) ⇒

- WAN is a types of Computer Network that Connects Computers over a Large geographical distance through a Shared Communication path.
- WAN Network Could be an interConnection between two or more LANs which are Connected through telephone lines or radio waves.



Advantages of WAN

- WAN allows Covering a Large geographical area.
- Multiple users Can share and access the internet at the Same time.
- High Bandwidth.

Disadvantages of WAN

- High initial investment Cost.

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- Hard to handle as the network is vast and complex
- Less secure.

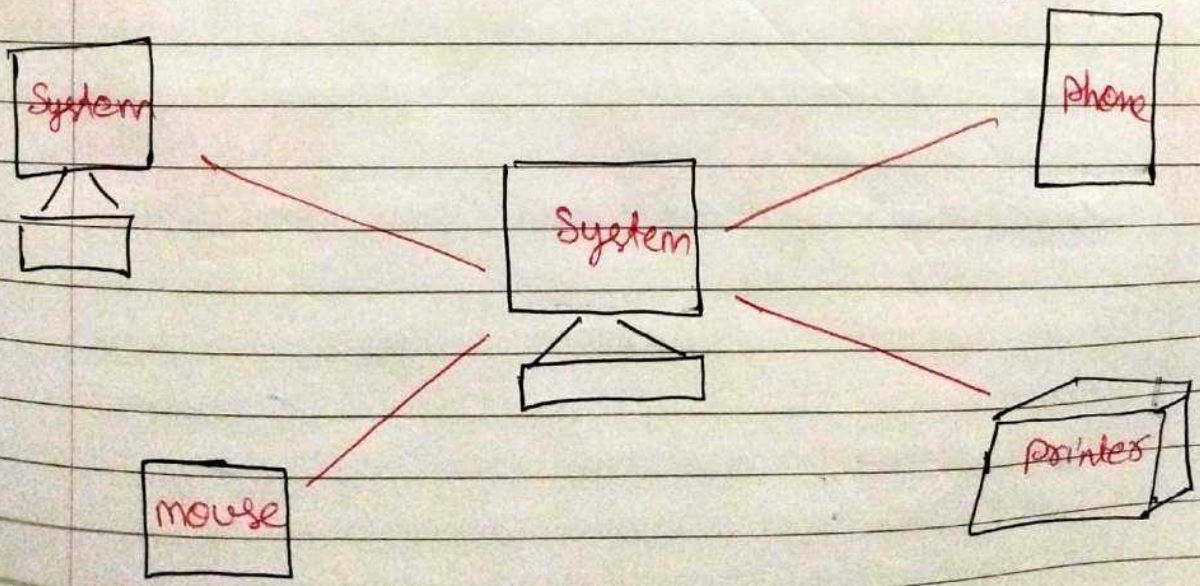
4) **PAN** (Personal Area Network) ⇒

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- It is the smallest and most basic type of computer network.

There are two types of PAN

1) Wired Personal Area Network :⇒ WPAN is created by using the USB.

2) Wireless Personal Area Network :⇒ WPAN is developed by simply using wireless technologies such as WiFi, Bluetooth. It is a low range network.



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Example of PAN are ⇒ USB, Computer, Bluetooth, etc.

Advantages of PAN are ⇒

- Less Expensive
- Confined to a small space area.
- Links to multiple devices concurrently.

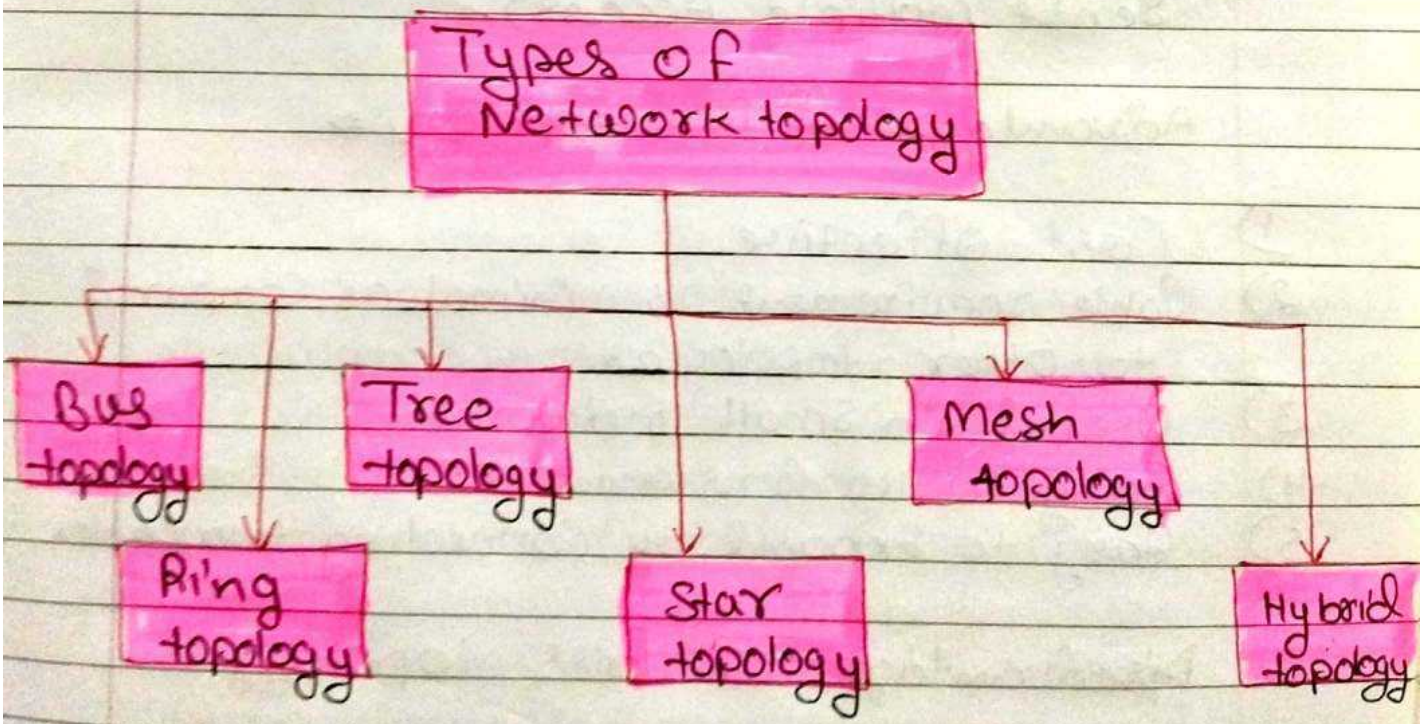
Disadvantages of PAN are ⇒

- Limited area
- Slow Data Transmission
- Interference with radio signal.

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Network Topology

- A Network Topology is the arrangement with which Computer System or network devices are connected to each other.
- The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as network topology.
- Topologies may define both physical and logical aspect of the network.
  Both logical and physical topologies could be same or different in a same network.



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1) Bus topology ⇒

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- The backbone cable is considered as a "single lane" through which the message is broadcast to all the stations.
- The most common access method of the bus topologies is CSMA (carrier Sense multiple ACCess).

Advantage of Bus topology ⇒

1) Cost-effective
2) Cable requirement is minimal as compared to other topologies
3) Useful in small networks
4) Easy to understand
5) Easy to Expand by Connecting two cables.

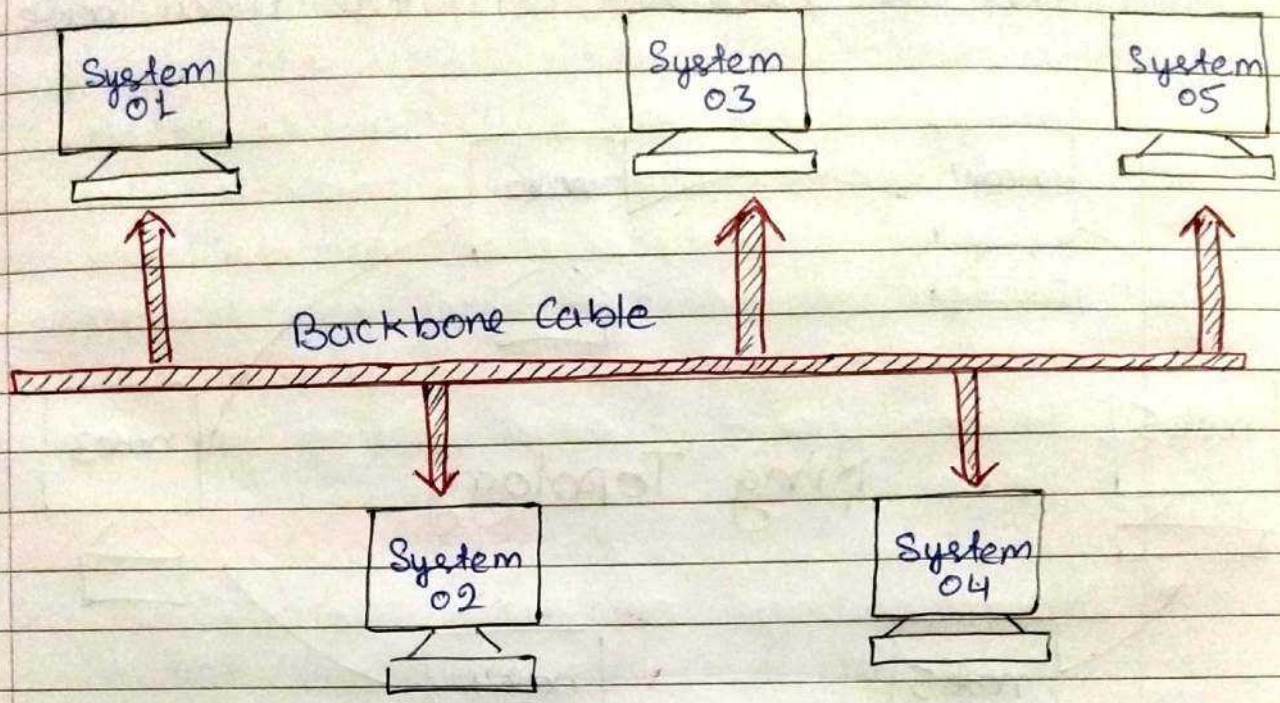
Disadvantage of Bus topology ⇒

1) The whole network fails if cables fail.
2) The performance of the network decreases in case network traffic is heavy or nodes are more or the cable has a

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limited length.

3) It works slower as compared to the ring topology.

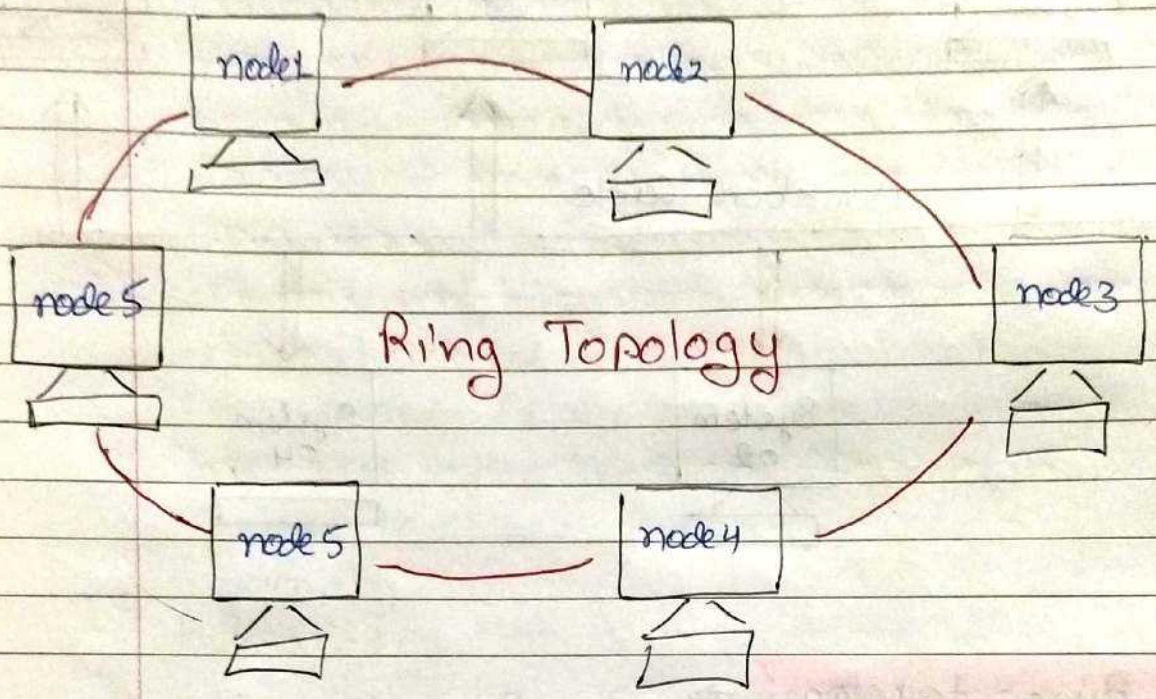


2) Ring topology ⇒

- It is named ring topology because it creates a ring as each computer is linked to the neighbouring computer, with the last one linked to the first, there are exactly two neighbours for each computer.
- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.

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- Data is transmitted in a sequential manner that is bit by bit. Data transmitted, has to go through each node linked in the network, till the final node.



Advantage of Ring Topology ⇒

1) Transmitting network is not affected by huge traffic or by the addition of more nodes, as only the nodes having tokens (short message) are allowed to transmit data.
2) Low Cost to install and Expand.
3) It is a more reliable network because the Communication System is not dependent on the single host Computer.
4) faulty devices can be removed from the network without bringing the network down.

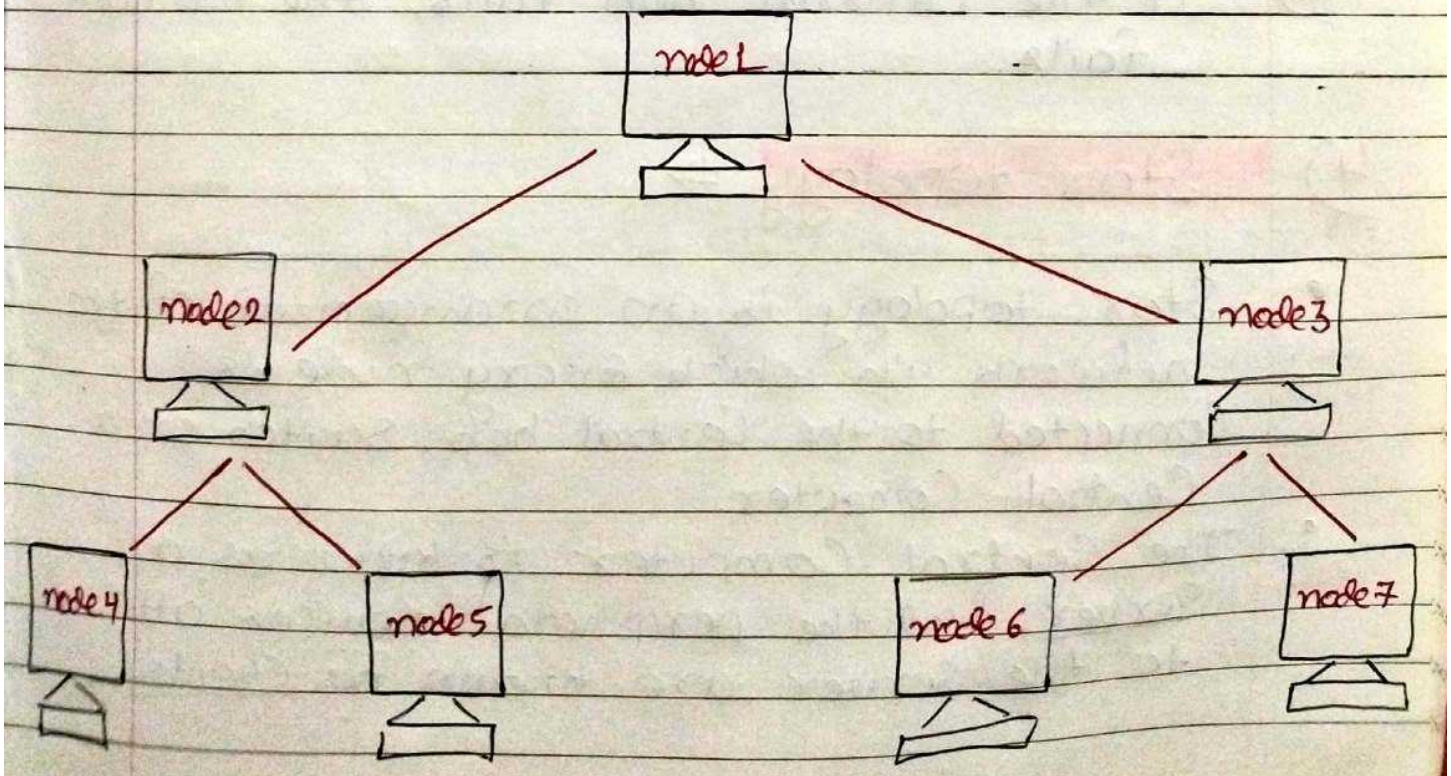
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Disadvantage of Ring topology ⇒

1) Troubleshooting is not simple in a ring topology.
2) The addition or removal of the computers interferes with the other nodes and network activity.
3) The crashing of one node affects the whole network.
4) Initial installation cost is high therefore not applied at low-density traffic.

3) Tree topology ⇒

- It has a root node and all other nodes are linked to it "creating a hierarchy."
- It is also called "hierarchical topology"
- It must have a minimum of three levels to the hierarchy.



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- Ideal if workstations are situated in groups
- Useful in Wide Area Network.

Advantage of Tree topology

1) Extension of bus and star topologies.
2) Expansion of nodes is possible and easy Easily managed and maintained.
3) Error detection and Error Correction are very easy.
4) It has point-to-point wiring for individual segments.

Disadvantage of tree topology

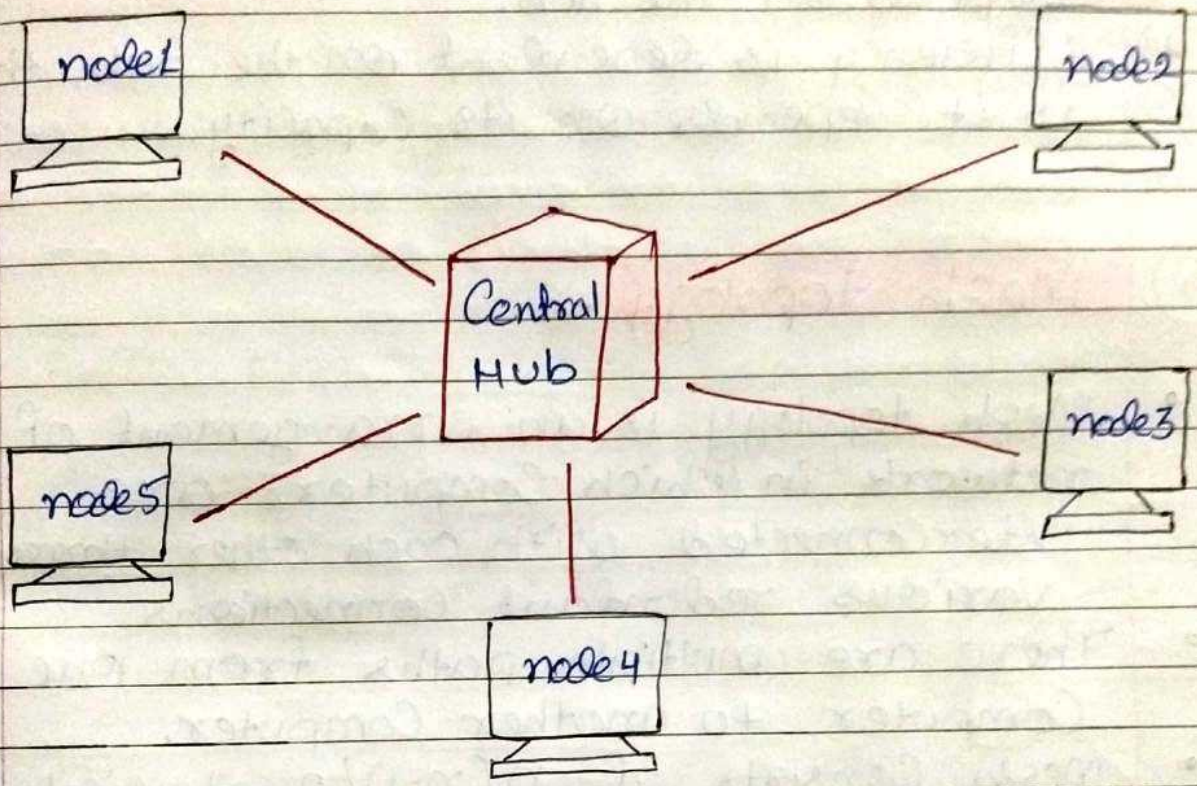
1) Heavily Cabled
2) Costly
3) If additional nodes are introduced, maintenance is difficult.
4) If the Central hub fails, the network fails.

4) Star topology ⇒

- Star topology is an arrangement of the network in which every node is Connected to the Central hub, Switch or a Central Computer.
- The Central Computer is known as a Server, and the peripheral devices attached to the Server are known as Clients.

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- Coaxial Cable or RJ-45 Cables are used to Connect the Computers.
- Hubs or Switches are mainly used as Connection devices in a physical star topology.
- Star topology is the most popular topology in network implementation.



Advantage of Star topology

1) Speedy performance with less number of nodes and low network traffic.
2) Hub can be easily upgraded without difficulty.
3) Simpler to troubleshoot.
4) Simpler to setup and modify.
5) If there is a failure in one node then the failed node is only affected, and the rest of the nodes can work without any issues.

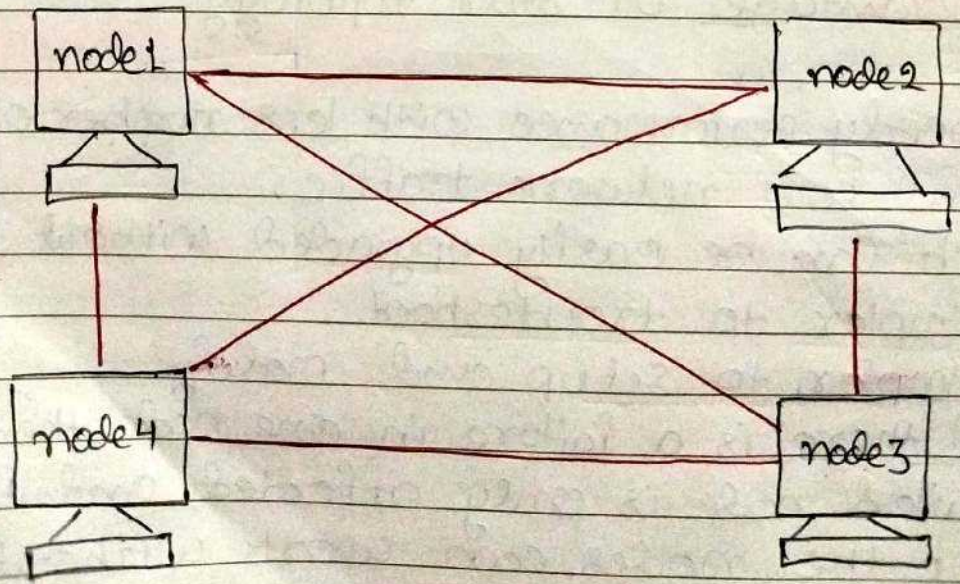
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Disadvantage of Star Topology

1) Expensive to install.
2) Expensive in usage.
3) If the hub crashes then the entire network is stopped because all linked nodes depend on the hub.
4) Efficiency is dependent on the hub, that is it depends on its capacity.

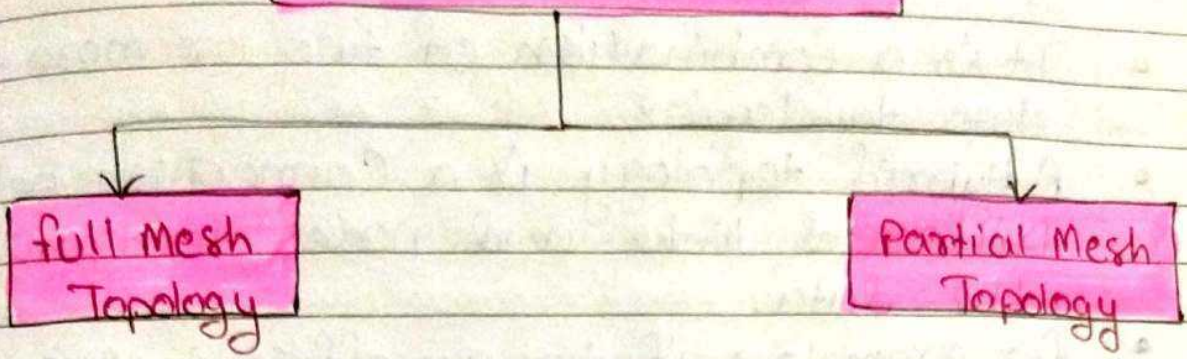
5) Mesh topology ⇒

- Mesh topology is an arrangement of the network in which Computers are interconnected with each other through various redundant connections.
- There are multiple paths from one Computer to another Computer.
- Mesh Consists of $n(n-1)/2$ physical Channels to link n number of devices.



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Types of Mesh topology



- full mesh Topology ⇒ In a full mesh topology, each Computer is connected to all the the Computers available in the network.

- Partial Mesh Topology ⇒ In a partial mesh topology, not all but certain Computers are connected to those Computers with which they Communicate frequently.

Advantage of mesh topology ⇒

1) Each Connection may carry its unique data load.
2) It is robust.
3) fault can be diagnosed easily.
4) Facilitates security and Privacy.

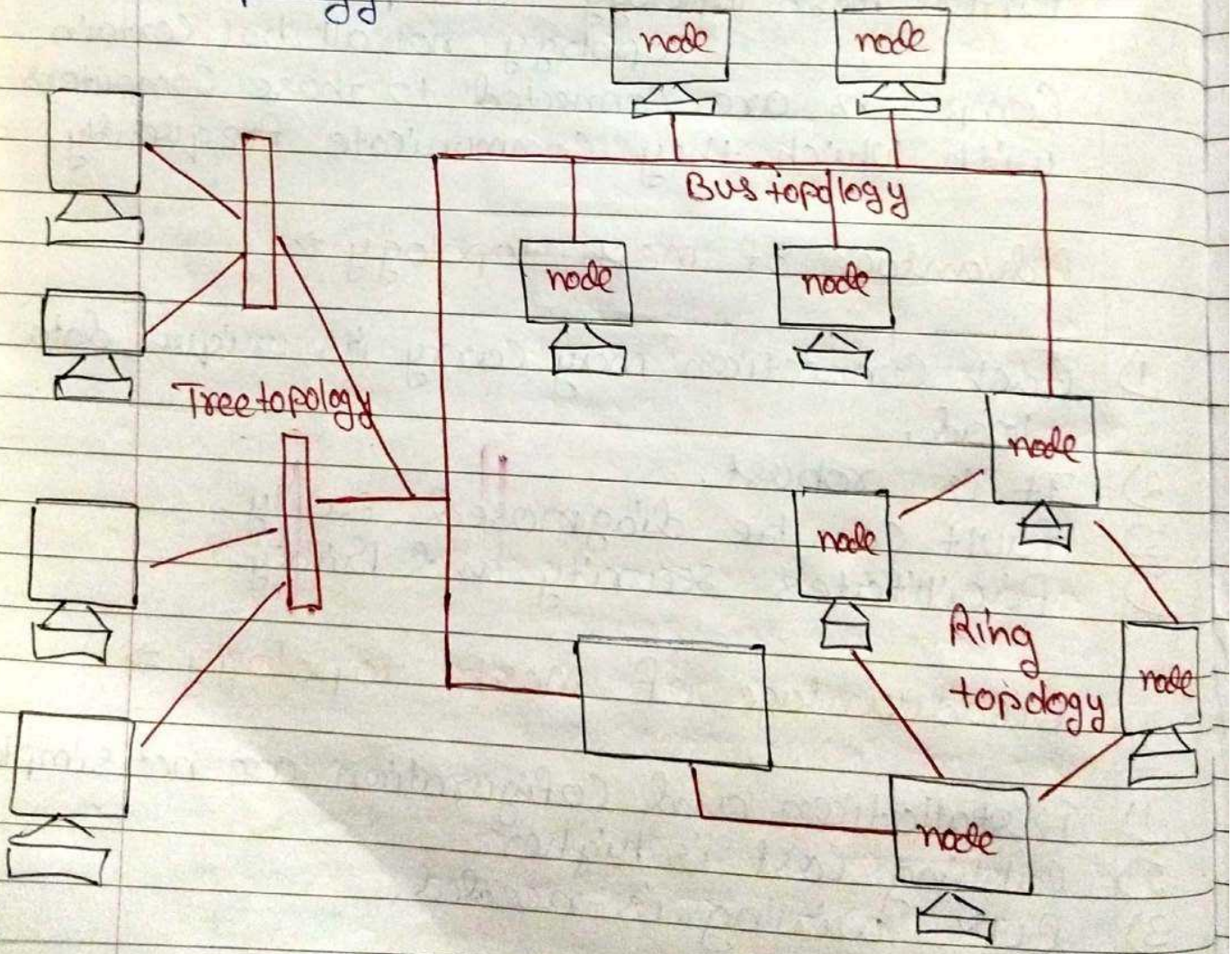
Disadvantage of Mesh topology ⇒

1) Installation and Cofiguration are not simple.
2) Cabling Cost is higher.
3) Bulk wiring is needed.

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6) Hybrid topology ⇒

- It is a Combination of two or more then two topologies.
- A Hybrid topology is a Connection between different links and nodes to transfer the data.
- for example ⇒ if in an office in some department ring topology is used and in another department in the same place, star topology is used, Connecting those topologies will form a Hybrid topology.



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Advantage of Hybrid topology

1) Reliable because error detecting and troubleshooting are easy.
2) Effective
3) Scalable as size can be increased easily
4) flexible.

Disadvantage of Hybrid Topology.

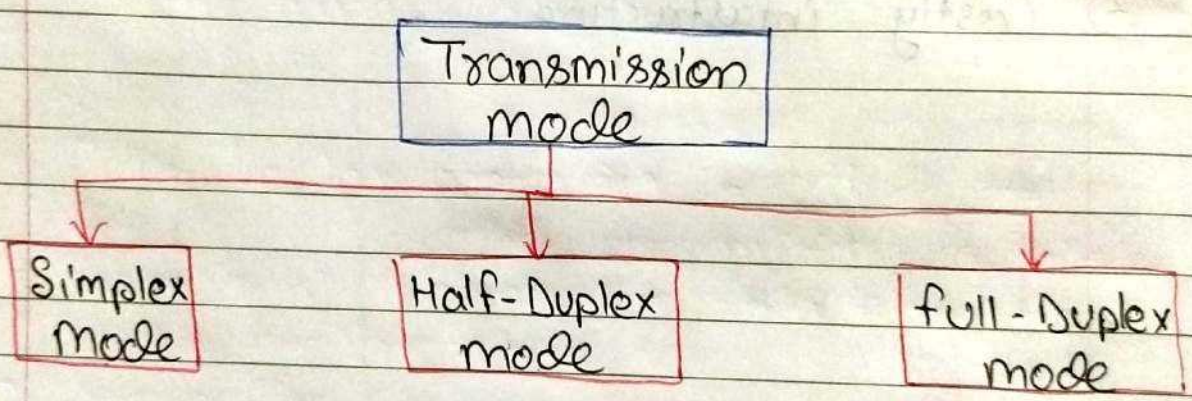
1) Complex in design
2) Costly Hub
3) Costly infastructure

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Transmission modes

- The way in which data is transmitted from one device to another device is known as transmission mode.
- Data Tranmission mode defines the direction of the flow of information between two Communication devices.
- The transmission mode is also known as the Communication mode. These modes direct the direction of flow of information.
- The transmission mode is defined in the physical Layer.

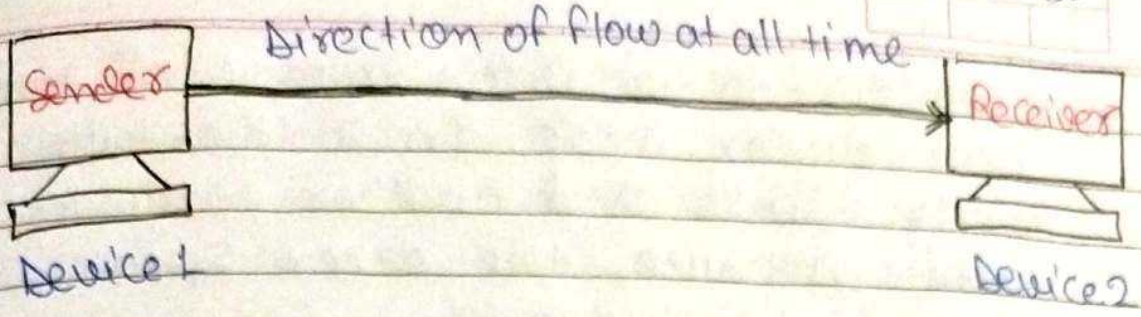
The transmission mode is devided into three Categories:



Simplex Mode ⇒ • Simplex is the data transmission mode in which the data can flow only in one direction, i.e. the communication is unidirectional.

- A device can only send the data but cannot receive it or it can receive the data but cannot send the data.

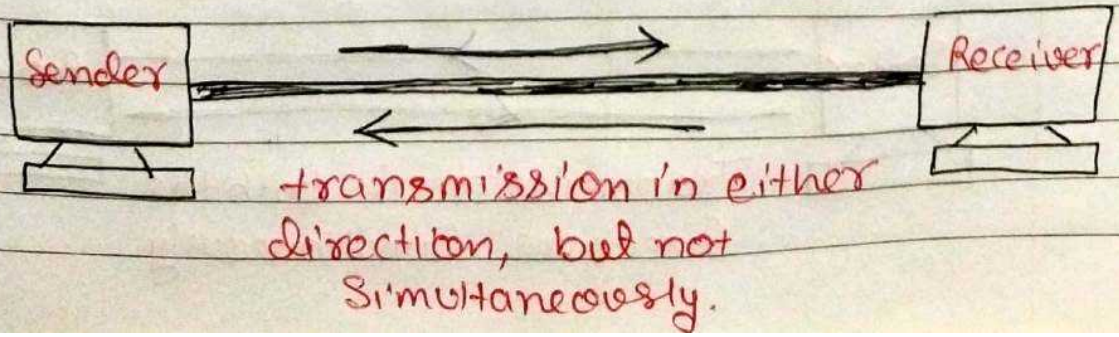
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- The Simplex mode is used in the business field as in Sales that do not require any corresponding reply.
- The radio station is a Simplex channel as it transmits the signal to the listeners but never allows them to transmit back.
- The main advantage of the Simplex mode is that the full capacity of the communication channel can be utilized during transmission.
- Communication is unidirectional, so it has no inter-communication between devices.

Half-Duplex Mode ⇒ • In half duplex channel, direction can be reversed. such as the station can transmit and receive the data as well.

- Message flow in both the directions, but not at the same time.
- In half-duplex mode, It is possible to perform the Error detection, and if any error occurs, then the receiver requests the sender to retransmit the data.

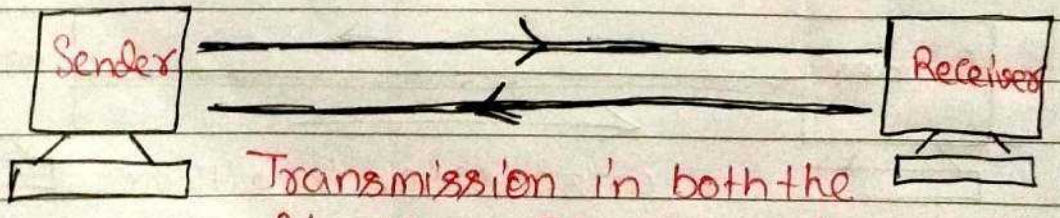


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- A Walkie-talkie is an example of the half-duplex mode. In walkie-talkie, one party speaks, and another party listen. After a pause, the other speaks and first party listens.
- main advantage of half-duplex mode, both the devices can send and receive the data and also can utilize the entire bandwidth of the communication channel during the transmission of data.
- One disadvantage of half-duplex mode, when one device is sending the data, then another has to wait, this causes the delay in sending the data at the right time.

Full-duplex mode ⇒ • In full duplex mode, the communication is bi-directional such as the data flow in both the directions.

- Both the stations can send and receive the message simultaneously.
- full-duplex mode has two simplex channels. One channel has traffic moving in one direction, and another channel has traffic flowing in the opposite direction.



Transmission in both the directions simultaneously

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- The full-duplex mode is the fastest mode of communication between devices.
- The most common example of the full duplex mode is a telephone network. When two people are communicating with each other by a telephone line, both can talk and listen at the same time.
- Advantage of Both the stations can send and receive the data at the same time.
- Disadvantage of full-duplex if there is no dedicated path exists between the devices, then the capacity of the communication channel is divided into two parts.

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Difference b/w Simplex, half duplex and full-duplex mode

Simplex mode	Half-duplex mode	full-duplex mode
1) the Communication is undirectional	1) the Communication is bidirectional, but one at a time.	1) The Communication is bidirectional
2) Sender and Receiver in Simplex mode, Sender can send the data but that Sender Cannot receive the data	2) Sender and Receiver in half duplex mode, Sender can send the data and also can receive the data but one at a time	2) Sender and Receiver in full duplex mode, Sender can send the data and also can receive the data Simultaneously
3) Usage of one Channel for the transmission of data	3) Usage of one Channel for the transmission of data	3) Usage of two channels for the transmission of data.
4) Simplex utilize the maximum of a Single bandwidth	4) the half-duplex involves lesser utilization of Single bandwidth at the time of transmission	4) the full-duplex doubles the utilization of transmission bandwidth.
5) Example of Simplex mode are: keyboard and monitor	5) Example of half duple mode is Walkie-talkies	5) Example of full-duple mode is: Telephone.

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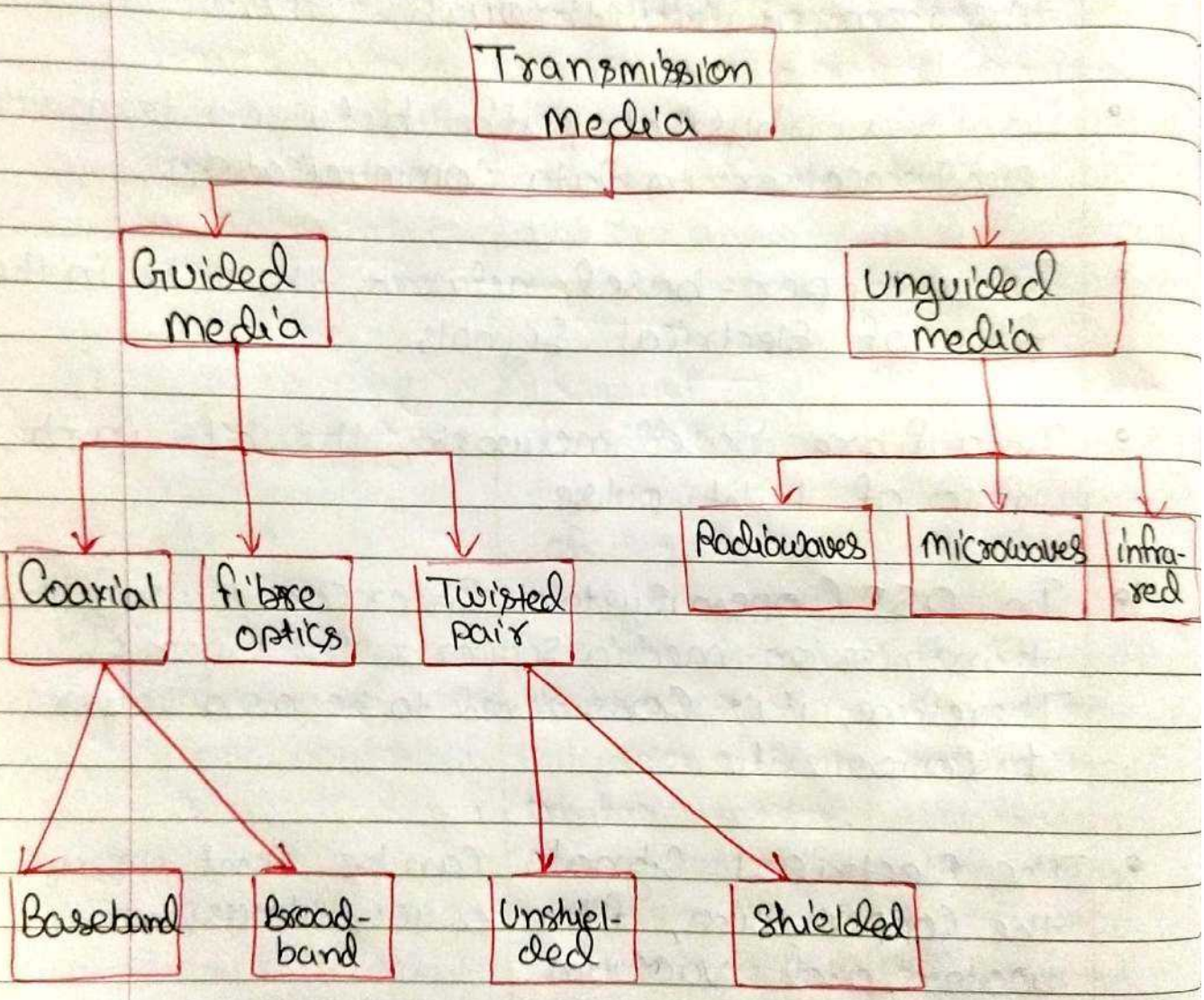
Transmission media

- Transmission media is a communication channel that carries the information from the sender to the receiver. Data is transmitted through the electromagenetic signals.
- The main functionality of the transmission media is to carry the information in the form of bits through LAN.
- It is a physical path between transmitter and receiver in data communication.
- In a copper-based network, the bits in the form of electrical signals.
- In a fibre based network, the bits in the form of light pulse.
- In OSI (open System Interconnection) phase transmission media supports the Layer 1. Therefore, it is considered to be as a Layer 1 Component.
- The Electrical signals can be sent through the copper wire, fibre optics, atmosphere, water and vaccum.
- Different transmission media have different properties such as bandwidth, delay, cost and ease of installation and maintenance.

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- The transmission media is available in the Lowest Layer of the OSI reference model ~~Physical Layer~~.

Classification of Transmission Media ⇒



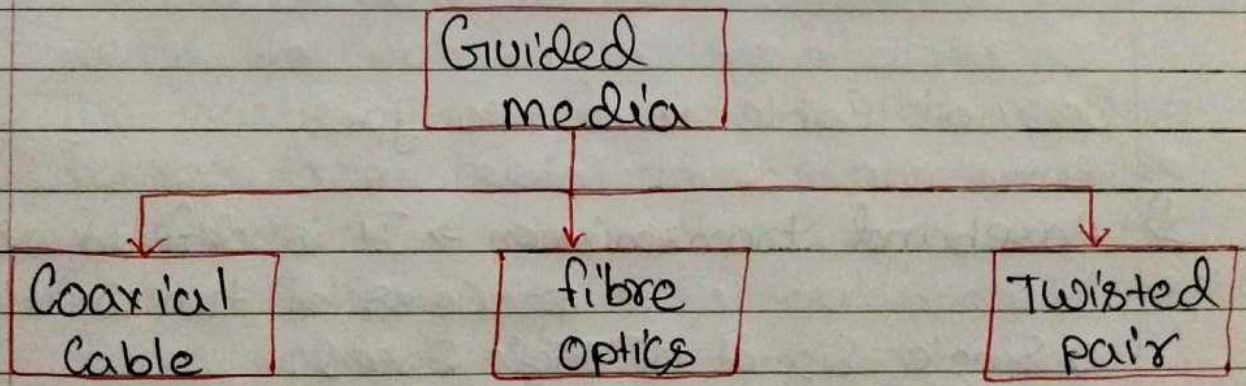
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Guided Transmission Media

- Guided media are also known as wired or bounded media.
- It is defined as the physical medium through which the signals are transmitted.
- Signals being transmitted are directed and confined in a narrow pathway by using physical links.

Feature of Guided media.

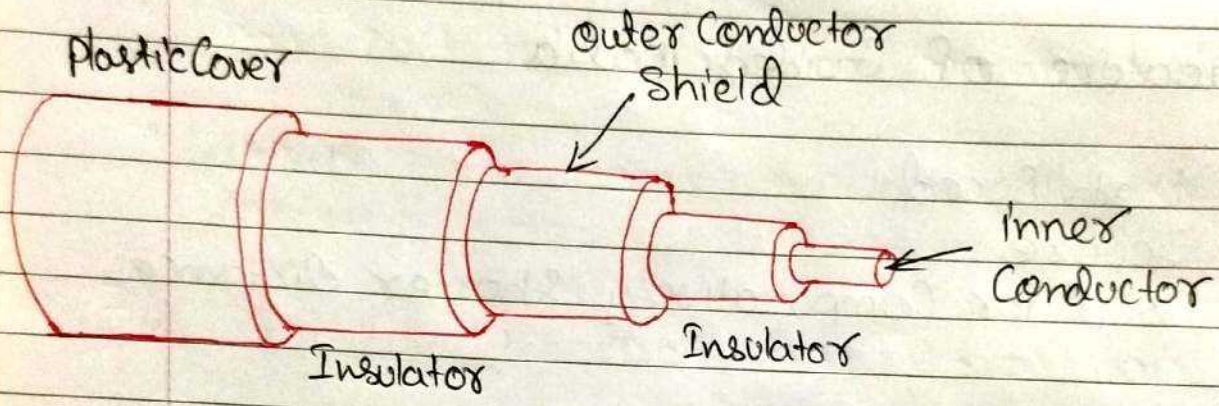
1) High Speed
2) Secure
3) Used for Comparatively shorter distance.



1) Coaxial Cable ⇒ • Coaxial Cable is very commonly used transmission media for example, TV wire is usually a Coaxial Cable.
- The name of the Cable is Coaxial as it Contains two Conductors parallel to each other.

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- It has a higher frequency as compared to Twisted pair cable.
- the inner Conductor of the Coaxial Cable is made up of Copper, and the outer Conductor is made up of copper mesh. the middle core is made up of non-Conductive Cover that Separates the inner Conductor from the outer Conductor.



Coaxial Cable is of two types:

1) Baseband transmission ⇒ It is defined as the process of transmitting a single Signal at high Speed.
2) Broadband transmission ⇒ It is defined as the process of transmitting multiple Signals Simultaneously.

Advantages of Coaxial Cable ⇒

- The data can be transmitted at high Speed.

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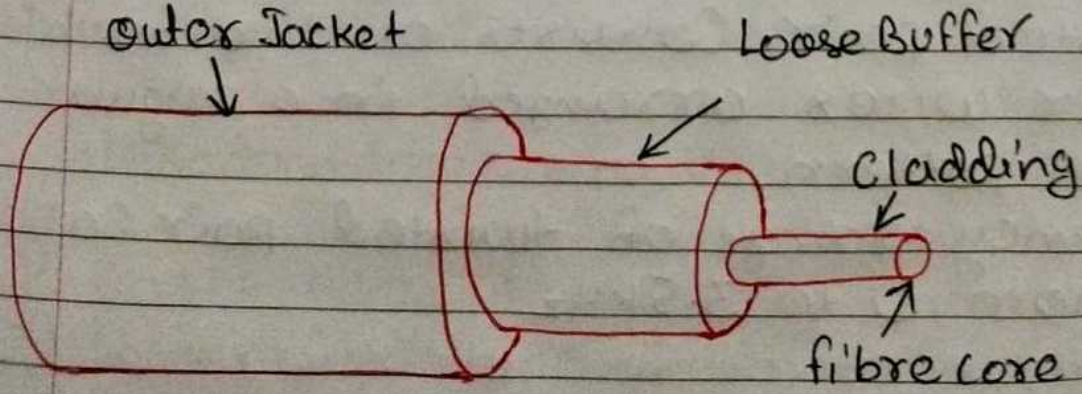
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages of Coaxial Cable ⇒

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

2) Fibre Optics ⇒ • fibre optic cable is a cable that uses electrical signals for communication.

- fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other type of wiring.
- fibre optics provide faster data transmission than copper wires.



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Advantage of fibre optics ⇒

- Increased Capacity and bandwidth
- Lightweight
- Less Signal attenuation
- Immunity to Electromagnetic interference
- Resistance to Corrosive materials.

Disadvantage of fibre optics ⇒

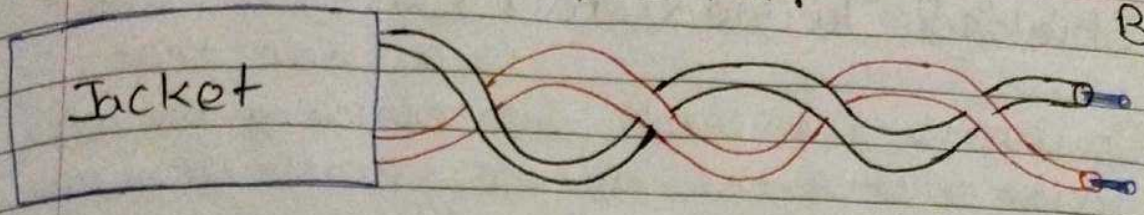
- Difficult to install and maintain
- High Cost
- fragile

3) Twisted Pair Cable ⇒ • Twisted pair is a physical media made up of a pair of Cables twisted with each other.

- A twisted pair Cable is cheap as Compared to other transmission media.
- Installation of the twisted pair Cable is easy, and it is a lightweight Cable.
- A wisted pair Consists of two insulated Copper wires arranged in a regular Spiral pattern.
- frequency range of twisted pair Cable is from 0 to 3.5KHz.

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Twisted Pair



Types of Twisted Pair

- Unshielded Twisted Pair
- Shielded Twisted Pair

- **Unshielded twisted Pair (UTP) ⇒** • UTP Consists of two insulated Copper wires twisted around one another.
- This type of cable has the ability to block interference and does not depend on a physical shield for this purpose.
- It is used for telephonic application.

Advantages of UTP ⇒

1) Least Expensive
2) Easy to install
3) High speed Capacity.

Disadvantages of UTP ⇒

1) Susceptible to External interference.
2) Lower Capacity and performance in comparison to STP.
3) Short distance transmission due to attenuation.

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- Shielded Twisted Pair (STP) ⇒ • STP is a cable that Contains the mesh surrounding the wire that allows the higher transmission rate.

Advantages of STP

1) Better performance at a higher data rate in Comparison to UTP.
2) Eliminates Crosstalk.
3) Comparatively faster.

Disadvantages of STP

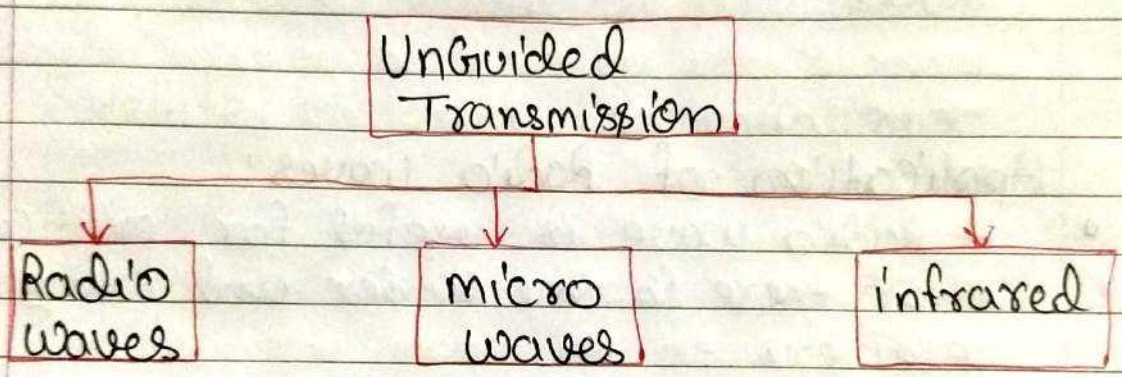
1) Comparatively difficult to install and manufacture.
2) More expensive
3) Bulky.

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UnGuided Transmission Media

- An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as Wireless transmission.
- In unguided media, air is the media through which the electromagnetic energy can flow easily.

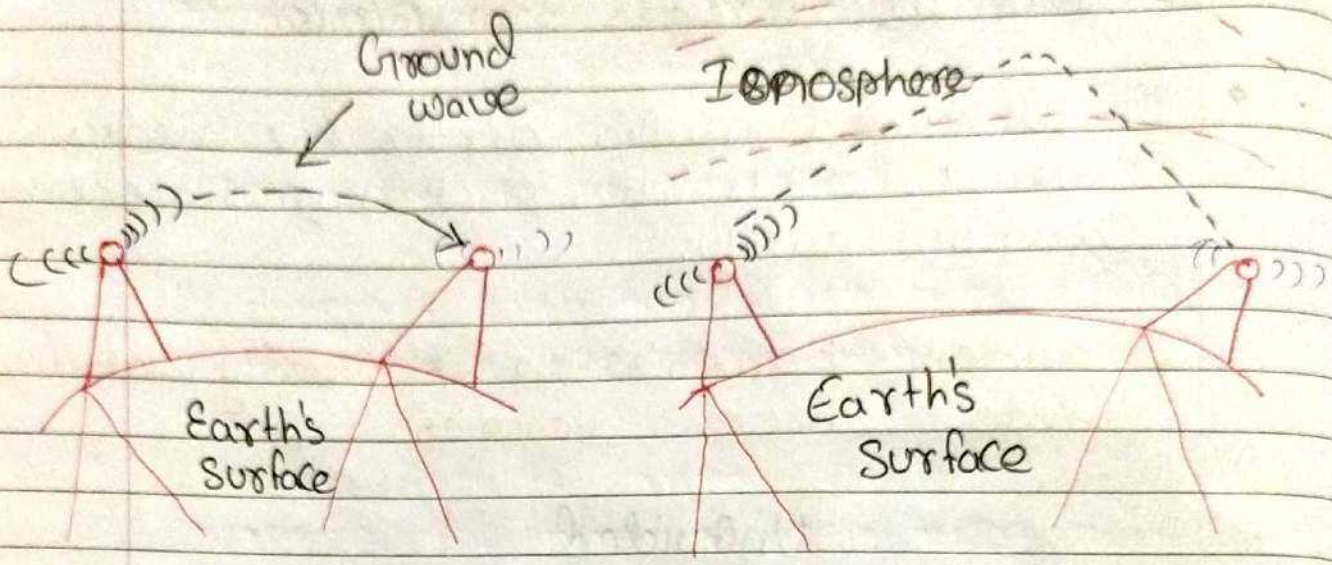
Unguided transmission is broadly classified into three categories:



Radio waves ⇒
- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves are omnidirectional such as the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1Khz.
- In the case of radio waves, the sending and receiving antenna are not aligned such that the wave sent by the sending antenna can be received by any receiving antenna.

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- An example of the radio wave is FM radio.



Application of Radio waves:

- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio, television, cordless phoes are example of a radio wave.

Microwaves ⇒ There are two types of microwaves transmission.

1) Terrestrial microwaves
2) Satellite microwave Communication.

1) Terrestrial Microwave Transmission ⇒

- Terrestrial microwave transmission is a

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- technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.

- Microwaves are the electromagnetic waves having the frequency in the range from 1GHz to 100GHz.

- Microwaves are unidirectional as the sending and receiving antenna is to be aligned such that the waves send by the sending antenna are narrowly focussed.

2) Satellite microwave Communication ⇒

- A Satellite is a physical object that revolves around the earth at a known height.

- Satellite Communication is more reliable nowadays as it offers more flexibility than cable and fibre optic systems.

- We can Communicate with any point on the globe by using Satellite Communication.

Infrared ⇒ • An infrared transmission is a wireless technology used for Communication over short ranges.

- The frequency of the infrared in the range from 300GHz to 400THz

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- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Characteristics of infrared ⇒

- It supports high bandwidth, and hence the data rate will be very high.
- Infrared waves cannot penetrate the walls. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.
- An Infrared communication provides better security with minimum interference.
- Infrared communication is unreliable outside the building because the sun rays will interfere with the infrared waves.

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OSI Model

- OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.

- OSI consists of seven layers, and each layer perform a particular network function.

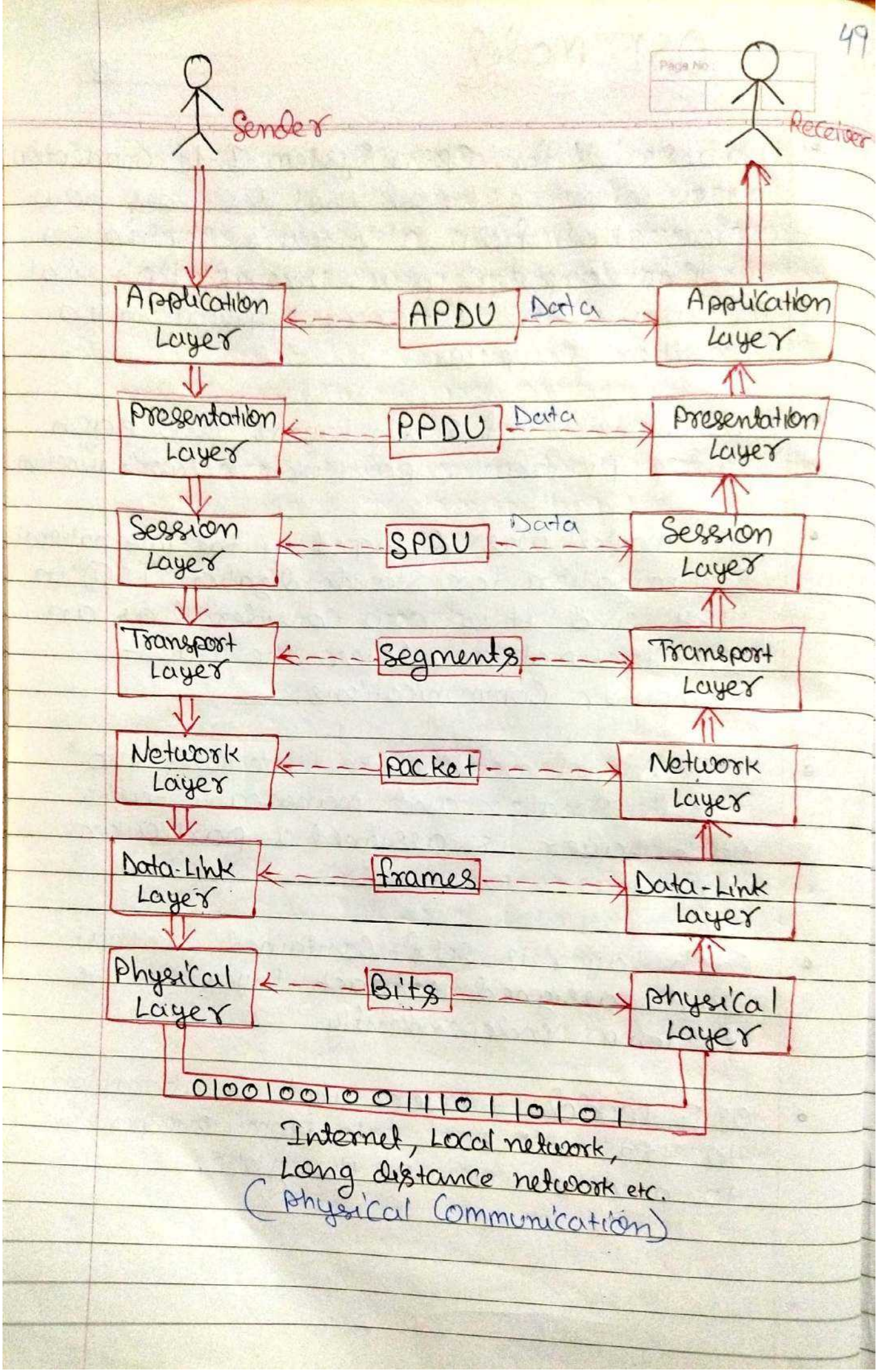
- OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.

- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.

- Each layer is self-contained, so that task assigned to each layer can be perform independently.

- OSI model 7 layers work collaboratively to transmit the data from one person to another across the globe.

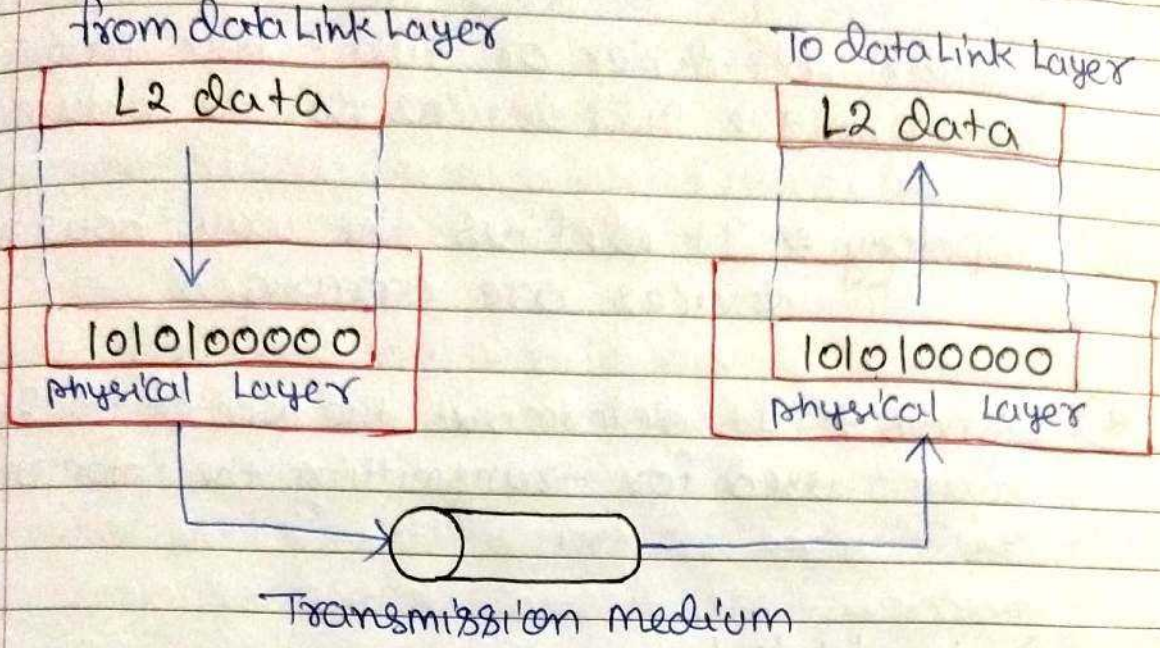
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1) Physical Layer ⇒



- The main functionality of the physical Layer is to transmit the individual bits from one node to another node.
- It is the Lowest Layer of the OSI model.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.

functions of Physical Layer ⇒

1) Line Configuration ⇒ It defines the way how two or more devices Can be Connected physically.

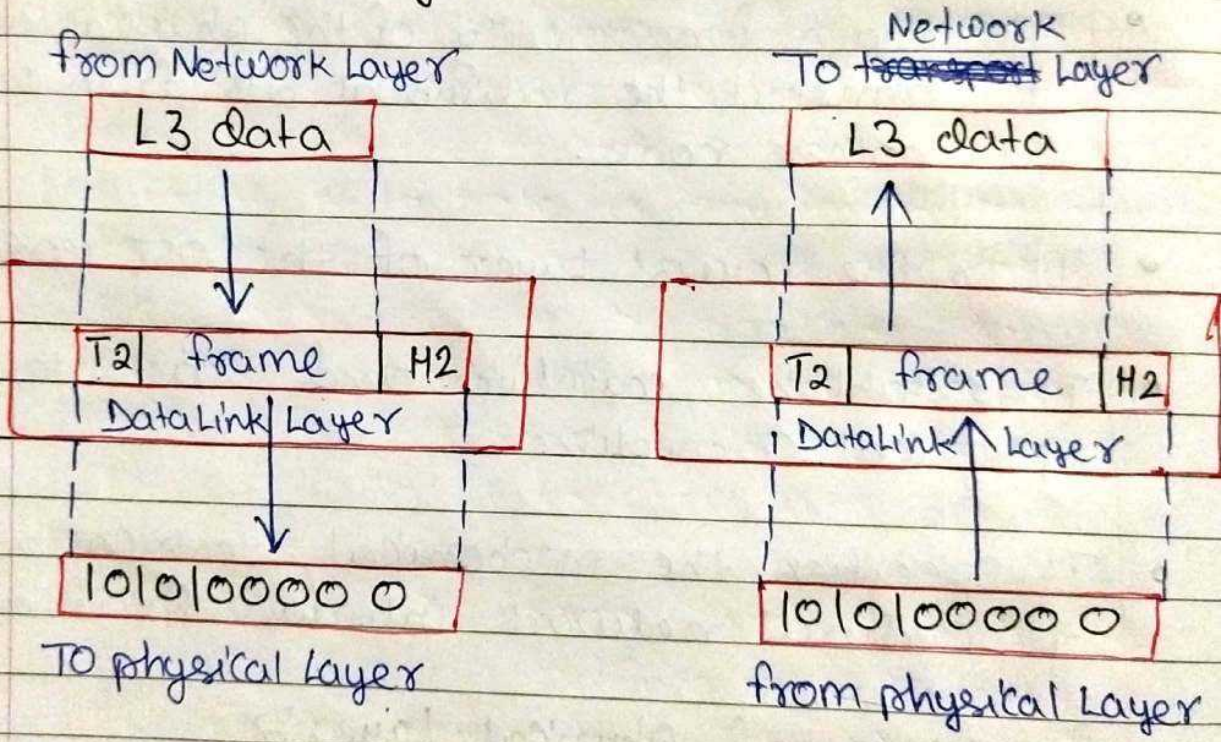
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2) Data transmission ⇒ It defines the transmission mode whether it is Simplex, half-duplex or full duplex mode between the two devices on the network.

3) Topology ⇒ It defines the way how network devices are arranged.

4) Signals ⇒ It determines the type of the signal used for transmitting the information.

2) Data-Link Layer ⇒



- This Layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.

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- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a Local network.
- It Contains two Sub Layers:

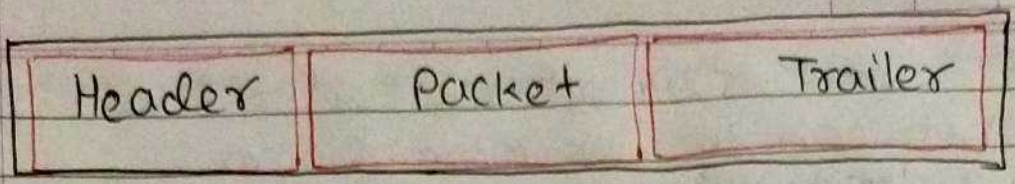
1) Logical Link Control Layer ⇒ • It is responsible for transferring the packets to the Network Layer of the receiver that is receiving.
- It identifies the address of the network Layer protocol from the header
- It provides flow control.

2) Media Access Control Layer ⇒ • A media access control layer is a link between the Logical Link Control Layer and the network's physical layer
- It is used for transferring the packets over the network.

functions of the Data-Link Layer ⇒

1) framing ⇒ The data Link layer translates the physical's raw bit stream into packets known as frames. The data Link layer adds the header and trailer to the frame.

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The header which is added to the frame Contains the hardware destination and source address.

2) Physical Addressing ⇒ The data Link Layer adds a header to the frame that Contains a destination address. The frame is transmitted to the destination address mentioned in the header.

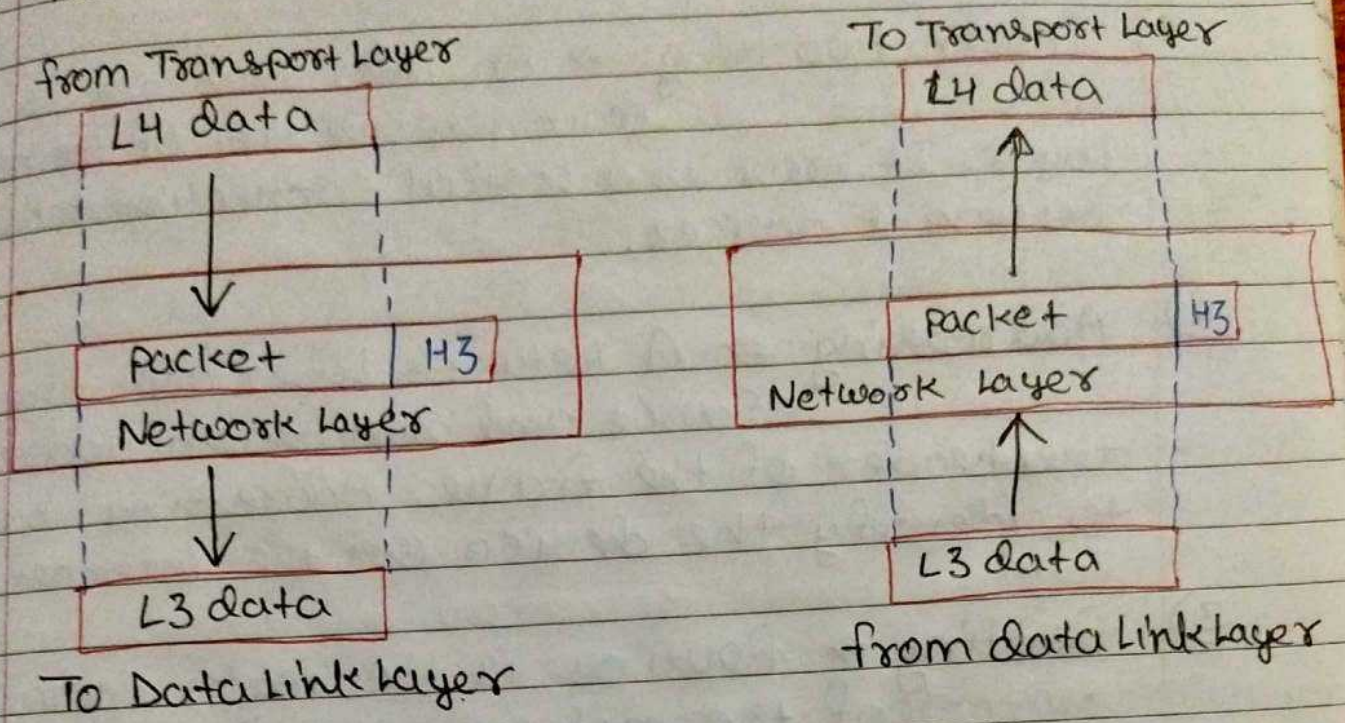
3) flow Control ⇒ flow Control is the main functionality of the Data-Link Layer. It is the technique through which the Constant data rate is maintained on both the Sides so that no data get ~~or~~ Corrupted. It ensures that the transmitting station such as a server with higher processing Speed does not Exceed the receiving Station, with Lower processing Speed.

4) Error Control ⇒ Error Control is achieved by adding a calculated value CRC (Cyclic Redundancy check) that is placed to the Data link Layer's trailer which is added to the message frame before it is sent to the physical Layer.

5) Access Control ⇒ When two or more devices are connected to the same Communication Channel, then the data Link Layer protocols are used to determine which device has control over the link at a given time.

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3) Network Layer ⇒



- It is a Layer 3 that manages device addressing, tracks the location of devices on the network.
- It determines the best path to move data from Source to the destination based on the network conditions, the priority of service and other factors.
- Data Link Layer is responsible for routing and forwarding the packets.
- Routers are the Layer 3 devices, they are Specified in this Layer and used to provide the routing services within an internetwork.
- The protocols used to route the network traffic are known as Network Layer protocols. Example of protocols are IP and IPv6.

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functions of Network Layer ⇒

1) Internetworking ⇒ An internetworking is the main responsibility of the network Layer. It provides a Logical Connection between different devices.

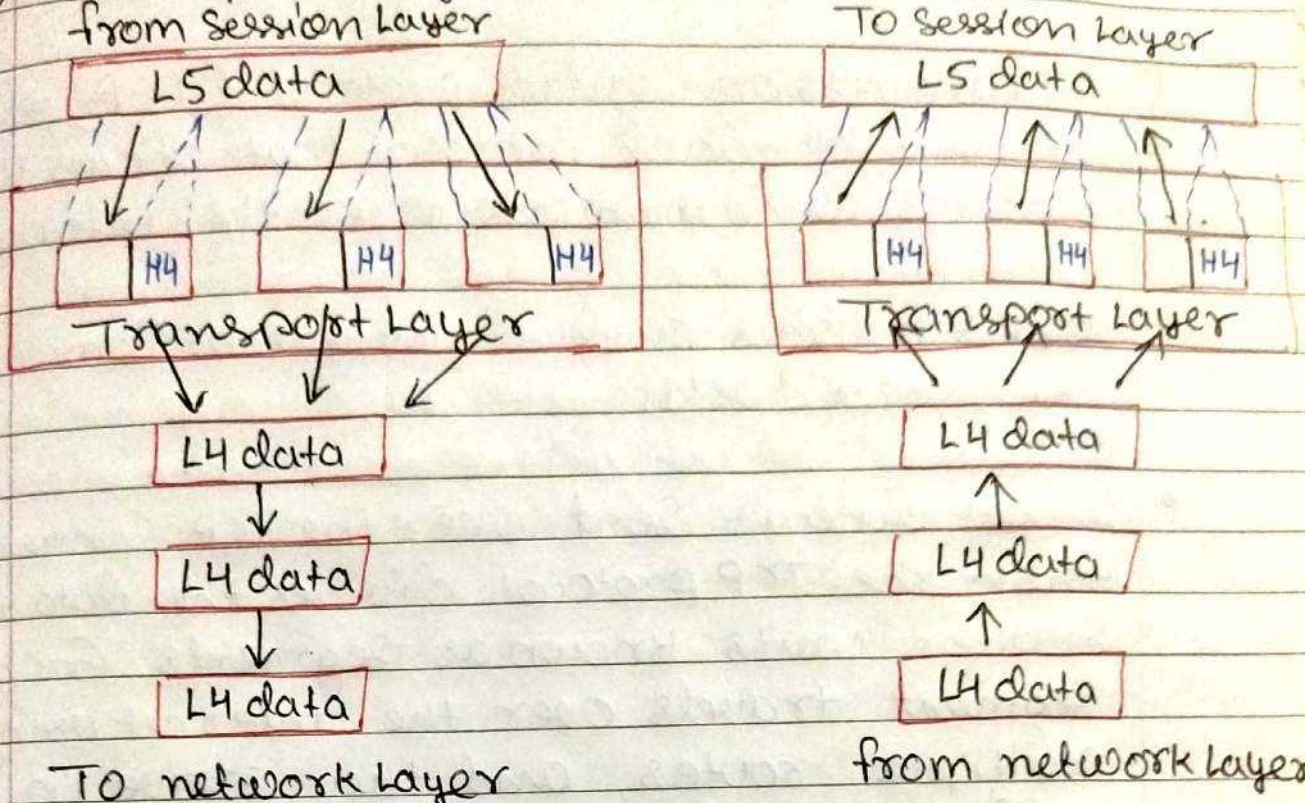
2) Addressing ⇒ A Network Layer adds the Source and destination address to the header of the frame. Addressing is used to identify the device on the internet.

3) Routing ⇒ Routing is the major component of the network Layer, and it determines the best optimal path out of the multiple path from source to destination.

4) Packetizing ⇒ A Network Layer receives the packets from the upper Layer and converts them into packets. This process is known as packetizing. It is achieved by internet protocol (IP).

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4) Transport Layer ⇒



- The trasport Layer is a Layer 4 ensure that messages are transmitted in the order in which they are send and there is no duplication of data.
- The main responsibility of the transport Layer is to transfer the data completely.
- It receives the data from the upper Layer and Converts them into smaller units known as segments.
- This Layer Can be termeds as an end-to-end Layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

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The two protocols used in this layer are! ⇒

1) Transmission Control protocol (TCP) ⇒

- It is a standard protocol that allows the systems to communicate over the internet.
- It establishes and maintains a connection between hosts.
- When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The TCP reorders the packets in the correct order at the receiving end.

2) User Datagram Protocol (UDP) ⇒

- User Datagram protocol is a transport layer protocol.
- It is an unreliable transport protocol as in this case receiver does not send any acknowledgement when the packet is received, the sender does not wait for any Acknowledgement. Therefore, this makes a protocol unreliable.

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functions of Transport Layer ⇒

1) service-point addressing ⇒ Computer run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one Computer to another Computer but also from one process to another process. The trasport Layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one Computer to another Computer and the responsibility of the transport Layer is to transmit the message to the Correct process.

2) Segmentation and reassembly ⇒ When the transport Layer receives the message from the upper Layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segments.

3) Connection Control ⇒ Transport Layer provides two services Connection-oriented service and Connectionless service. A Connectionless service treats each segments as an individual packet, and they all travel in different routes to reach the destination.

4) flow Control ⇒ The transport Layer also reponsible for ~~error~~ flow Control.

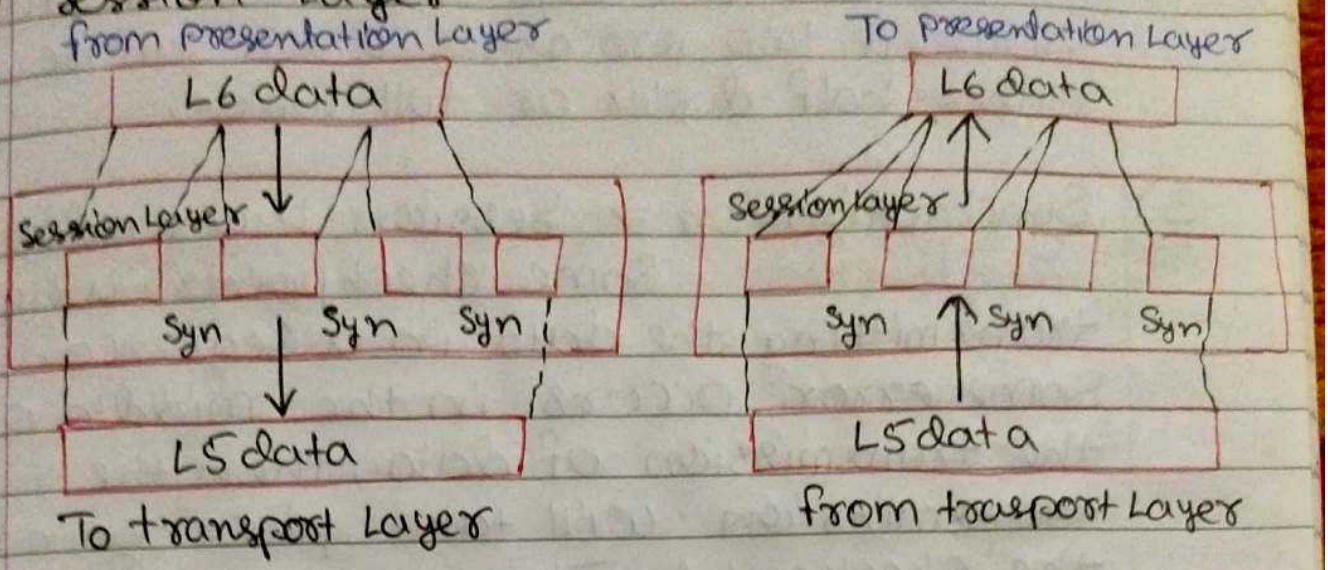
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but it is performed end-to-end rather than across a single link.

5) Error Control ⇒ The transport layer is also responsible for Error Control. Error Control is performed end-to-end rather than across the single link. The sender transport layer ensure that message reach at the destination without any error.

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5) Session Layer



- It is a Layer 3 in the OSI model.
- The session Layer is used to establish, maintain and synchronizes the interaction between communicating devices.
- This Layer is responsible for the establishment of connection, maintenance of sessions, authentication, and also ensures security.

functions of session Layer:

1) Session establishment, maintenance and termination :⇒ The Layer allows the two processes to establish, use the terminate a connection.

2) Dialog Control ⇒ Session Layer acts as a dialog Controller that Creates a dialog between two processes or we

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Can say that it allows the Communication between two processes which can be either half-duplex or full-duplex.

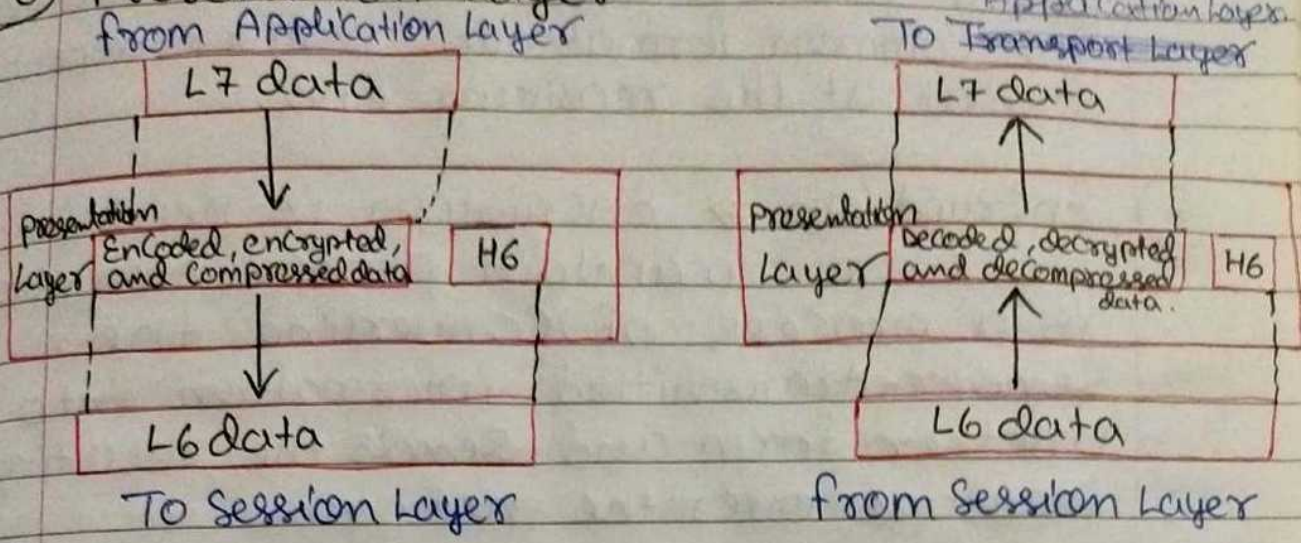
3) Synchronization ⇒ Session Layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

msg ⇒ Hello P.T

Key +3 KHOOR Encode C.T

-3 HELLO → P.T

6) Presentation Layer



- A Presentation Layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It acts as a data translator for a network.
- This Layer is a part of the operating system that converts the data from one presentation format to another format.
- The Presentation Layer is also known as the Syntax Layer.

functions of Presentation Layer ⇒

1) Translation ⇒ The processes in two systems exchange the information in the form of character string, number and so on.
It converts the data from sender-dependent

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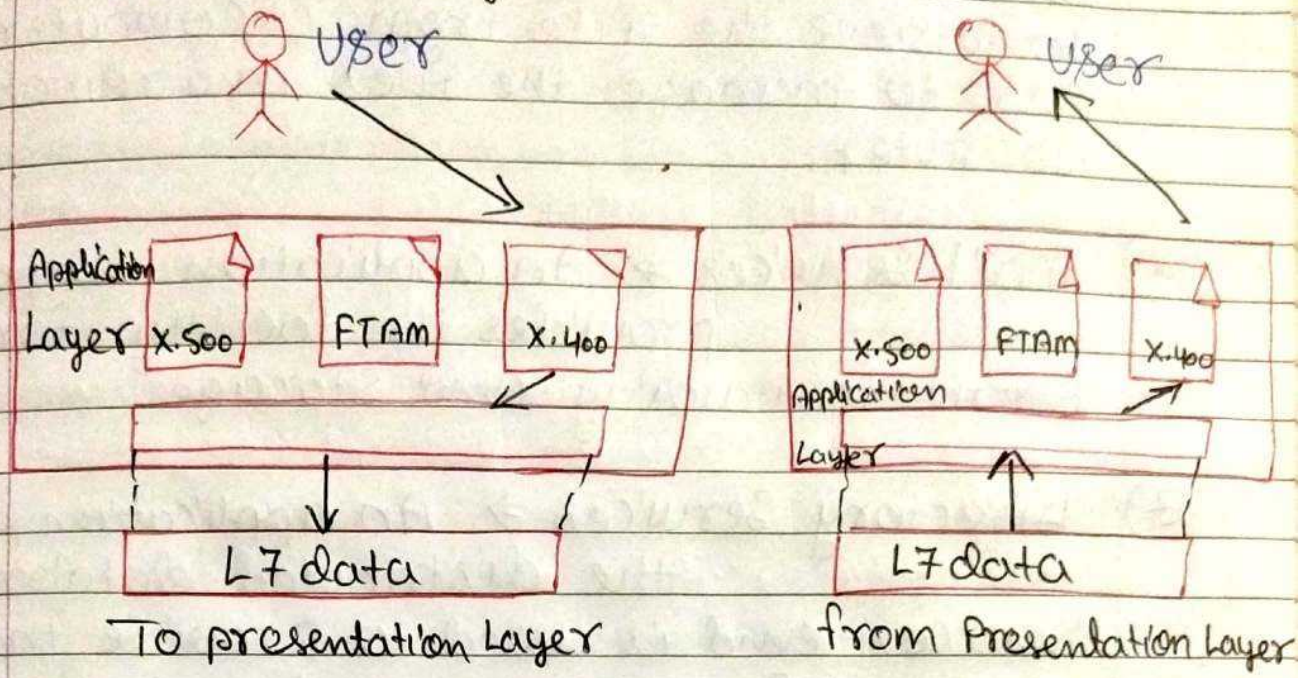
format into a common format and changes the common format into receiver-dependent format at the receiving end.

2) Encryption ⇒ Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.

3) Compression ⇒ Data compression is a process of compressing the data, such that it reduce the no. of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

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7) Application Layer ⇒



- An application Layer serves as a window for users and application process to access network service.
- It handles issues such as network transparency, resource allocation, etc.
- An application Layer is not an application, but it performs the application Layer function.
- This Layer provides the network services to the end-users.

function of Application Layer ⇒

1) file transfer, access, and management (FTAM) ⇒ An application Layer allows a user

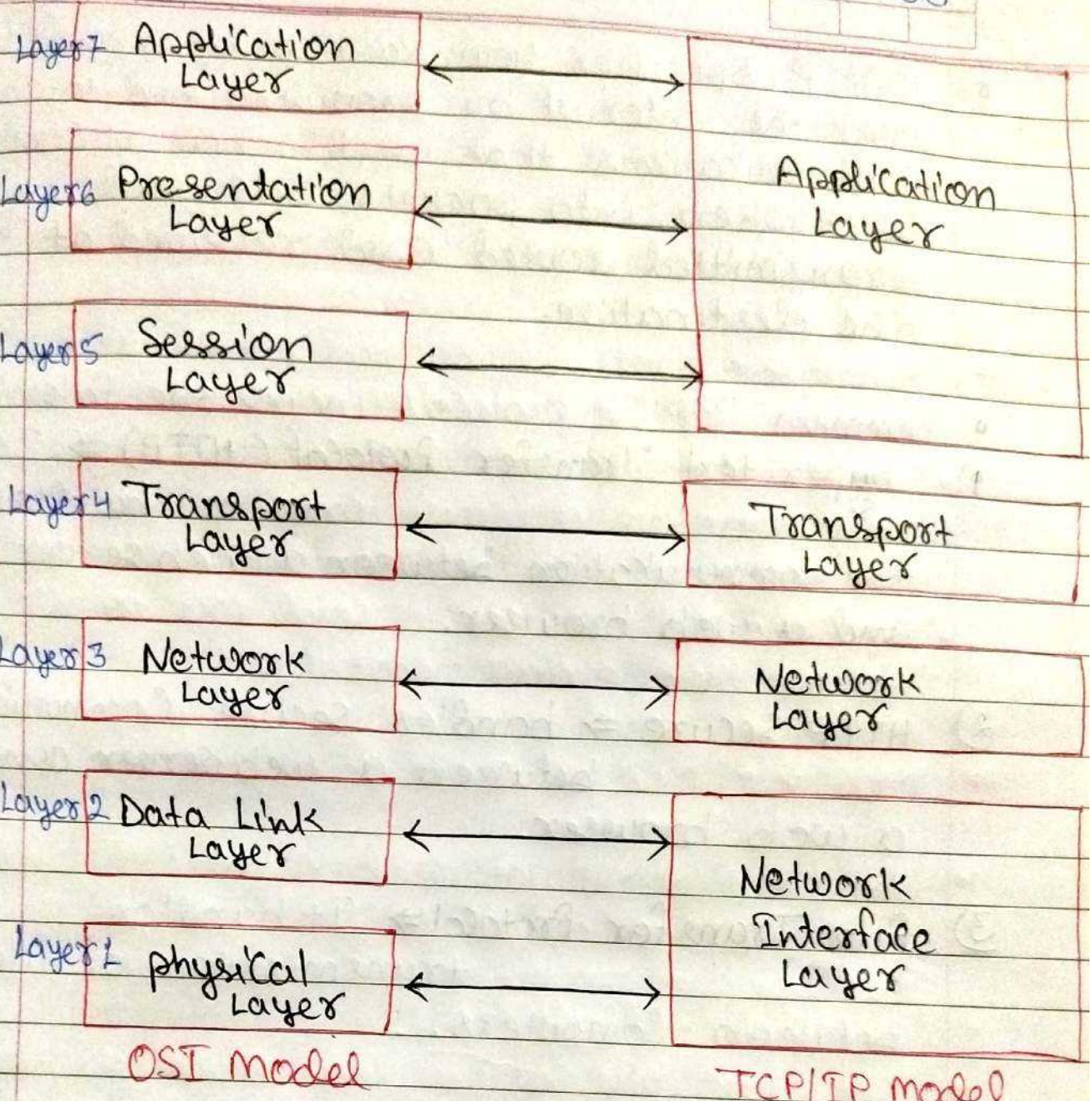
to access the files in a remote Computer, to retrieve the file from a Computer and to manage the files in a remote Computer.

2) Mail services ⇒ An application Layer provides the facility for email forwarding and storage.

3) Directory services ⇒ An application provides the distributed database sources and is used to provide that global information about various objects.

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TCP/IP Model



- TCP/IP Stands for Transmission Control Protocol / Internet Protocol and it's a suite of Communication protocols used to interconnect network devices on the internet.
- TCP/IP it's also used as a Communication protocol in a private Computer network (an internet and Extranet).

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- TCP/IP Specifies how data is exchanged over the internet by providing end-to-end Communications that identify how it should be broken into packets, addressed, transmitted routed and received at the destination.

- Common TCP/IP protocols include the following:

1) Hyper text Transfer Protocol (HTTP) ⇒ It handles the communication between a web server and a web browser.

2) HTTP Secure ⇒ handles secure Communication between a web server and a web browser.

3) File Transfer Protocol ⇒ It handles transmission of files between Computers.

- The TCP/IP model is a Concise version of the OSI model. It Contains four Layers, Unlike seven Layers in the OSI model. The Layer are:

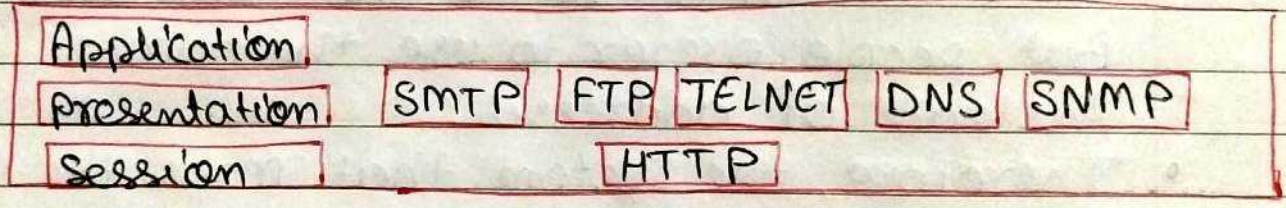
1) Application Layer / process
2) Transport Layer / Host-to-Host
3) Internet Layer / Network Layer
4) Network Interface / Link Layer

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1) Application Layer ⇒

- An application Layer is the topmost Layer in the TCP/IP model.
- It is responsible for handling high-Level protocols, issues of representation.
- This Layer allows the user to interact with the application.
- When one application Layer protocol wants to communicate with another application layer, it forwards its data to the transport Layer.
- Example of the application Layer is an application such as file transfer, email, remote login etc.

function and protocols of Application Layer ⇒



- HTTP ⇒ • This protocol allows us to access the data over the world wide web.
- It transfer the data in the form of plain text, audio, video.
- It is known as a hypertext transfer protocol as it has the efficiency to use in a hypertext environment where there are rapid jumps from one document to another.

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- **SNMP ⇒** • SNMP stands for Simple Network management Protocol.
- It is a framework used for managing the devices on the internet by using the TCP/IP protocol suite.

- **SMTP ⇒** • SMTP stands for Simple mail Transfer protocol.
- The TCP/IP protocol that support the e-mail is known as a Simple mail transfer protocol.
- This protocol is used to send the data to another e-mail address.

- **DNS ⇒** • DNS stands for Domain Name System
- An IP address is used to identify the Connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses.
- Therefore, the system that maps the Name to the address is known as Domain Name System.

- **TELNET ⇒** • It stands for Terminal Network or teletype Network Protocol.
- It establishes the Connection between the Local Computer and remote Computer in such a way that the local terminal appears to be a terminal at the remote System.

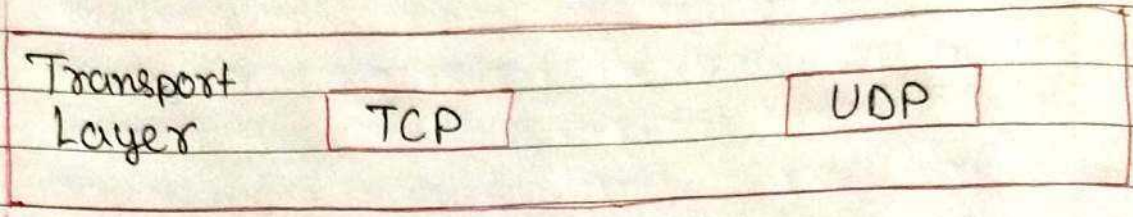
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- FTP ⇒ • FTP stands for file transfer Protocol.
- FTP is a standard internet protocol used for transmitting the files from one computer to another computer.

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2) Transport Layer ⇒

- the transport Layer is responsible for the reliability, flow Control, and Correction of data which is being sent over the network.



The two protocols used in the transport Layer are User Datagram protocol and Transmission Control protocol.

User Datagram Protocol (UDP) ⇒

- It provides Connectionless service and end-to-end delivery of transmission.
- It is an unreliable protocol as it discovers the error but not Specify the error.
- User Datagram Protocol discovers the error, and ICMP protocol reports the error to the sender that user datagram has been damaged.

ICMP (Internet Control message Protocol)

⇒ UDP Consists of the following fields:

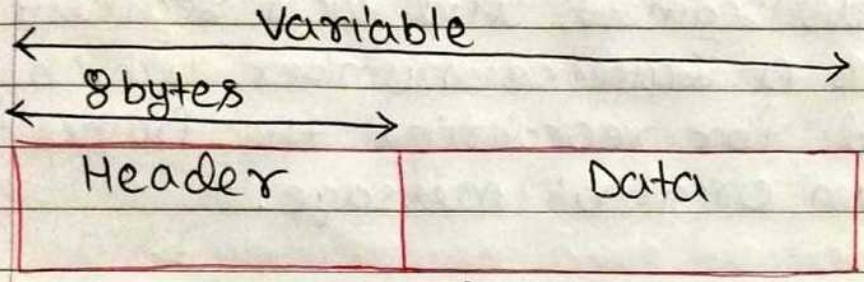
Source port address ⇒ The source port address is the address of the

application program that has created the message.

Destination port address ⇒ The destination port address is the address of the application program that receives the message.

Total Length ⇒ It defines the total no. of bytes of the user datagram in bytes.

Checksum ⇒ The checksum is a 16-bit field used in error detection.



Header format

Source port address 16 bit	Destination port address 16 bit
Total length 16 bits	Checksum 16 bits

Transmission Control Protocol (TCP) ⇒

- It provides a full transport Layer services to applications.
- It creates a virtual circuit between the sender and receiver, and it is active for

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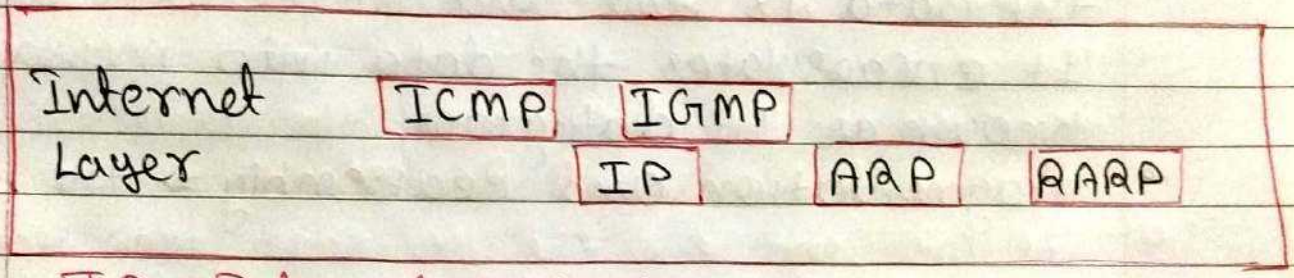
the duration of the transmission.

- TCP is a reliable protocol as it detects the Error and retransmits the damaged frames. Therefore, it ensures all the segments must be received and acknowledged before the transmission is considered to be Completed and a virtual Circuit is discarded.
- At the sending end, TCP divides the whole message into smaller units known as segment, and each segment Contains a sequence number which is required for reordering the frames to form an original message.
- At the receiving end, TCP Collects all the segments and reorders them based on sequence numbers.

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3) Internet Layer ⇒

- An Internet Layer is the second Layer of the TCP/IP model.
- An internet Layer is also known as the network Layer.
- The main responsibility of the internet Layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.



IP ⇒ Internet protocol
ARP ⇒ Address Resolution protocol
ICMP ⇒ Internet Control Message protocol
IGMP ⇒ Internet Group Management Protocol
RARP ⇒ Reverse Address Resolution Protocol.

following are the protocols used in this Layer are; ⇒

1) IP Protocol ⇒ IP protocol is used in this Layer, and it is the most significant part of the entire TCP/IP suite.

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following are the responsibilities of this protocol:

- IP addressing ⇒ this protocol implements Logical host addresses known as IP address.
- Host-to-host communication ⇒ It determines the path through which the data is to be transmitted.
- Data Encapsulation and formatting ⇒ An IP protocol accepts the data from the transport layer protocol. An IP protocol ensures that the data is sent and received securely, It encapsulates the data into message known as IP datagram.
- fragmentation and Reassembly ⇒ the limit imposed on the size of the IP datagram by data link Layer protocol is known as maximum Transmission unit (MTU).
  If the size of IP datagram is greater then the MTU unit, then the IP protocol splits the datagram into smaller units so that they can travel over the Local network.
- Routing ⇒ When IP datagram is sent over the same local network such as LAN, MAN, WAN, It is known as direct delivery. When source and destination are on the distant network, then the IP datagram is sent indirectly.

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2) ARP Protocol ⇒

- ARP stands for Address Resolution Protocol.
- ARP is a network Layer protocol which is used to find the physical address from the IP address.

The two terms are mainly associated with the ARP protocol:

- ARP request ⇒ When a sender wants to know the physical address of the device, it broadcasts the ARP request to the network.

- ARP reply ⇒ Every device attached to the network will accept the ARP request and process the request, but only recipient recognize the IP address and sends back its physical address in the form of ARP reply.

The recipient adds the physical address both to its cache memory and to the datagram header.

3) ICMP Protocol ⇒

- ICMP stands for Internet Control Message Protocol.
- It is a mechanism used by the hosts or routers to send notifications regarding datagram problems back to the sender.

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- A datagram travels from router-to-router unit it reaches its destination.

- An ICMP protocol mainly uses two terms:
  ICMP test ⇒ ICMP test is used to test whether the destination is reachable or not.
  ICMP Reply ⇒ ICMP Reply is used to check whether the destination device is responding or not.

- The core responsibility of the ICMP protocol is to report the problems, not correct them. The responsibility of the correction lies with the sender.

- ICMP can send the messages only to the source, but not to the intermediate routers because the IP datagram carries the address of the source and destination but not of the router that it is passed to.

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4) Network Interface (Access) Layer ⇒

- A Network Layer is the Lowest Layer of the TCP/IP model.
- A Network Layer is the combination of the physical Layer and Data link Layer defined in the OSI reference model.
- It defines how the data should be sent physically through the network.
- This Layer is mainly responsible for the transmission of the data between two devices on the same network.
- The function carried out by this Layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP address into physical addresses.
- The protocols used by this Layer are ethernet, token ring, FDDI, X.25, frame relay.

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Difference between OSI and TCP/IP Model

OSI Model	TCP/IP Model
1) OSI Stands for Open System Interconnection	1) TCP/IP implies Transmission Control Protocol / Internet Protocol.
2) The OSI model was developed by ISO (International Standard organization) in 1984.	2) The TCP/IP model was developed by ARPANET (Advanced Research Project Agency Network) in 1982.
3) It Consists of 7 Layers: Starting from the bottom they are the physical, data link, Network, Transport, session, Presentation and Application Layer.	3) It Consists of 4 Layers: Starting from the bottom they are the Network Interface, Internet, Transport and Application Layer.
4) The OSI model follows a vertical approach.	4) The TCP/IP model follows a horizontal approach.
5) In the OSI model, the transport Layer provides a guarantee for the delivery of the packets.	5) In TCP/IP transport Layer does not provide the surety for the delivery of packets. But still, we can say that It is a reliable model.

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OSI model	TCP/IP model
6) In the OSI model, the Physical Layer and data Link Layer are Seprate Layer.	6) In TCP/IP, Physical and data Link Layers are merged as a single network Layer.
7) In OSI model, the Session and presentation Layers are seprated such as both the Layers are Different.	7) In this model, the Session and presentation Layer are not different Layers. Both Layers are included in the application Layer.

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